# Table 2: Coverage (%) − Shrinkage

Scenario I: all means are 0 for T0,T1,...,T10

| | None adjustment | | | | | | | | | | |
|---|---|---|---|---|---|---|---|---|---|---|---|
| **Missing rate** | **T1vs.T0** | **T2vs.T0** | **T3vs.T0** | **T4vs.T0** | **T5vs.T0** | **T6vs.T0** | **T7vs.T0** | **T8vs.T0** | **T9vs.T0** | **T10vs.T0** | **Joint** |
| 0% | 95.05 | 94.71 | 94.69 | 95.12 | 94.77 | 95.24 | 94.91 | 94.98 | 94.89 | 95.16 | 72.79 |
| **Imputation using linear regression** | | | | | | | | | | | |
| 10% | 95.27 | 94.62 | 94.36 | 94.53 | 94.79 | 94.63 | 94.40 | 94.73 | 94.61 | 94.71 | 71.41 |
| 20% | 95.23 | 94.77 | 94.90 | 95.24 | 95.26 | 94.80 | 95.09 | 95.17 | 94.94 | 94.81 | 72.19 |
| 30% | 95.31 | 95.60 | 95.12 | 95.32 | 95.14 | 95.54 | 95.51 | 95.36 | 95.69 | 95.19 | 73.89 |
| 40% | 95.72 | 95.62 | 96.01 | 95.87 | 95.53 | 95.98 | 95.63 | 95.88 | 95.77 | 95.79 | 75.81 |
| **Imputation using predictive mean matching method** | | | | | | | | | | | |
| 10% | 94.28 | 93.58 | 93.46 | 93.61 | 93.90 | 93.64 | 93.38 | 93.78 | 93.84 | 93.75 | 67.40 |
| 20% | 92.97 | 92.76 | 92.43 | 92.60 | 93.02 | 92.45 | 92.62 | 92.91 | 92.65 | 92.68 | 63.02 |
| 30% | 91.14 | 91.47 | 91.02 | 91.35 | 91.06 | 91.62 | 91.12 | 91.22 | 91.72 | 91.11 | 57.60 |
| 40% | 89.64 | 89.36 | 89.86 | 89.79 | 89.60 | 89.59 | 89.76 | 89.60 | 89.64 | 89.41 | 51.42 |
| **Imputation using propensity−score method** | | | | | | | | | | | |
| 10% | 94.30 | 93.49 | 93.48 | 93.47 | 93.90 | 93.68 | 93.23 | 93.80 | 93.64 | 93.60 | 67.12 |
| 20% | 92.85 | 92.41 | 92.23 | 92.78 | 92.91 | 92.11 | 92.63 | 92.79 | 92.61 | 92.37 | 62.10 |
| 30% | 90.93 | 91.09 | 90.69 | 91.00 | 90.60 | 91.23 | 90.93 | 90.82 | 91.31 | 91.04 | 56.43 |
| 40% | 89.06 | 89.12 | 88.89 | 88.79 | 88.69 | 89.17 | 89.08 | 89.34 | 88.92 | 88.97 | 49.43 |

Requested replicates per setting: 10000 | Imputations: 40 | Successful sample sizes: completion_summary.csv

# Table 2: Coverage (%) − Shrinkage

Scenario I: all means are 0 for T0,T1,...,T10

| | Many−to−one adjustment | | | | | | | | | | |
|---|---|---|---|---|---|---|---|---|---|---|---|
| **Missing rate** | **T1vs.T0** | **T2vs.T0** | **T3vs.T0** | **T4vs.T0** | **T5vs.T0** | **T6vs.T0** | **T7vs.T0** | **T8vs.T0** | **T9vs.T0** | **T10vs.T0** | **Joint** |
| 0% | 99.31 | 99.24 | 99.34 | 99.29 | 99.14 | 99.16 | 99.17 | 99.15 | 99.00 | 99.17 | 94.38 |
| **Imputation using linear regression** | | | | | | | | | | | |
| 10% | 99.47 | 99.30 | 99.32 | 99.15 | 99.30 | 99.21 | 99.07 | 99.16 | 99.16 | 99.22 | 94.56 |
| 20% | 99.47 | 99.40 | 99.35 | 99.48 | 99.43 | 99.23 | 99.38 | 99.41 | 99.18 | 99.31 | 95.18 |
| 30% | 99.47 | 99.57 | 99.36 | 99.40 | 99.33 | 99.45 | 99.50 | 99.41 | 99.47 | 99.43 | 95.78 |
| 40% | 99.54 | 99.45 | 99.50 | 99.47 | 99.43 | 99.54 | 99.56 | 99.51 | 99.40 | 99.52 | 95.89 |
| **Imputation using predictive mean matching method** | | | | | | | | | | | |
| 10% | 99.26 | 98.97 | 99.00 | 98.98 | 98.96 | 99.05 | 98.74 | 98.85 | 98.84 | 98.91 | 92.69 |
| 20% | 98.93 | 98.85 | 98.79 | 98.83 | 98.90 | 98.54 | 98.81 | 98.83 | 98.51 | 98.67 | 91.46 |
| 30% | 98.28 | 98.41 | 98.25 | 98.23 | 97.94 | 98.36 | 98.13 | 98.21 | 98.23 | 98.20 | 88.59 |
| 40% | 97.59 | 97.60 | 97.76 | 97.51 | 97.30 | 97.53 | 97.48 | 97.61 | 97.14 | 97.51 | 84.31 |
| **Imputation using propensity−score method** | | | | | | | | | | | |
| 10% | 99.28 | 99.02 | 98.99 | 98.99 | 98.97 | 99.00 | 98.74 | 98.90 | 98.89 | 98.93 | 92.85 |
| 20% | 98.84 | 98.75 | 98.75 | 98.87 | 98.84 | 98.45 | 98.78 | 98.66 | 98.33 | 98.49 | 90.85 |
| 30% | 98.35 | 98.49 | 97.94 | 97.95 | 98.00 | 98.13 | 98.16 | 98.08 | 98.11 | 97.77 | 87.88 |
| 40% | 97.34 | 97.37 | 97.52 | 97.13 | 97.03 | 97.04 | 97.24 | 97.36 | 96.80 | 97.11 | 82.82 |

Requested replicates per setting: 10000 | Imputations: 40 | Successful sample sizes: completion_summary.csv

# Table 2: Coverage (%) − Shrinkage

Scenario II: mu0=0, mu1=1.5, mu2=2.0, ..., mu10=6.0

| | None adjustment | | | | | | | | | | |
|---|---|---|---|---|---|---|---|---|---|---|---|
| **Missing rate** | **T1vs.T0** | **T2vs.T0** | **T3vs.T0** | **T4vs.T0** | **T5vs.T0** | **T6vs.T0** | **T7vs.T0** | **T8vs.T0** | **T9vs.T0** | **T10vs.T0** | **Joint** |
| 0% | 94.98 | 95.11 | 95.25 | 95.36 | 95.05 | 94.76 | 95.06 | 95.46 | 94.70 | 95.03 | 73.01 |
| **Imputation using linear regression** | | | | | | | | | | | |
| 10% | 94.78 | 95.16 | 94.74 | 94.18 | 94.48 | 94.79 | 94.44 | 94.54 | 94.95 | 94.56 | 70.36 |
| 20% | 95.24 | 94.68 | 95.31 | 95.30 | 95.20 | 95.16 | 95.24 | 95.47 | 95.10 | 95.03 | 73.21 |
| 30% | 95.01 | 95.74 | 95.50 | 95.12 | 95.33 | 95.36 | 95.37 | 95.42 | 95.30 | 95.42 | 74.20 |
| 40% | 96.00 | 95.77 | 95.71 | 96.04 | 95.86 | 95.87 | 95.60 | 95.94 | 95.84 | 95.62 | 76.07 |
| **Imputation using predictive mean matching method** | | | | | | | | | | | |
| 10% | 93.85 | 94.00 | 93.77 | 93.07 | 93.28 | 93.67 | 93.56 | 93.73 | 93.84 | 93.48 | 66.07 |
| 20% | 93.28 | 92.64 | 92.99 | 92.98 | 93.04 | 92.97 | 92.93 | 93.02 | 92.85 | 92.82 | 63.72 |
| 30% | 91.49 | 91.77 | 91.43 | 91.62 | 91.43 | 91.30 | 91.73 | 91.71 | 91.45 | 91.48 | 58.77 |
| 40% | 89.38 | 89.24 | 88.86 | 89.35 | 89.15 | 89.25 | 89.41 | 89.58 | 89.51 | 89.31 | 50.35 |
| **Imputation using propensity−score method** | | | | | | | | | | | |
| 10% | 93.77 | 94.03 | 93.58 | 93.22 | 93.11 | 93.68 | 93.41 | 93.70 | 93.79 | 93.46 | 65.67 |
| 20% | 93.16 | 92.37 | 92.79 | 92.67 | 92.62 | 92.61 | 92.80 | 92.71 | 92.42 | 92.46 | 62.64 |
| 30% | 90.83 | 91.40 | 91.03 | 90.98 | 90.60 | 90.48 | 90.76 | 90.71 | 90.54 | 90.79 | 55.58 |
| 40% | 89.11 | 88.93 | 88.41 | 88.42 | 88.68 | 88.52 | 88.03 | 88.42 | 87.95 | 87.56 | 46.68 |

Requested replicates per setting: 10000 | Imputations: 40 | Successful sample sizes: completion_summary.csv

# Table 2: Coverage (%) − Shrinkage

Scenario II: mu0=0, mu1=1.5, mu2=2.0, ..., mu10=6.0

| | Many−to−one adjustment | | | | | | | | | | |
|---|---|---|---|---|---|---|---|---|---|---|---|
| **Missing rate** | **T1vs.T0** | **T2vs.T0** | **T3vs.T0** | **T4vs.T0** | **T5vs.T0** | **T6vs.T0** | **T7vs.T0** | **T8vs.T0** | **T9vs.T0** | **T10vs.T0** | **Joint** |
| 0% | 99.28 | 99.28 | 99.28 | 99.31 | 99.20 | 99.12 | 99.19 | 99.31 | 98.94 | 99.32 | 94.29 |
| **Imputation using linear regression** | | | | | | | | | | | |
| 10% | 99.45 | 99.35 | 99.38 | 99.25 | 99.22 | 99.28 | 99.25 | 99.26 | 99.32 | 99.15 | 94.91 |
| 20% | 99.45 | 99.46 | 99.48 | 99.22 | 99.33 | 99.32 | 99.27 | 99.37 | 99.31 | 99.08 | 95.43 |
| 30% | 99.34 | 99.42 | 99.42 | 99.43 | 99.42 | 99.39 | 99.30 | 99.27 | 99.36 | 99.37 | 95.44 |
| 40% | 99.48 | 99.54 | 99.46 | 99.40 | 99.46 | 99.54 | 99.48 | 99.43 | 99.52 | 99.40 | 96.32 |
| **Imputation using predictive mean matching method** | | | | | | | | | | | |
| 10% | 99.24 | 99.07 | 99.10 | 98.90 | 98.94 | 99.11 | 99.07 | 99.03 | 98.99 | 98.85 | 93.12 |
| 20% | 98.82 | 98.71 | 98.85 | 98.76 | 98.64 | 98.68 | 98.59 | 98.72 | 98.64 | 98.49 | 91.60 |
| 30% | 98.30 | 98.42 | 98.51 | 98.48 | 98.41 | 98.11 | 98.23 | 98.17 | 98.10 | 98.16 | 88.75 |
| 40% | 97.63 | 97.61 | 97.41 | 97.32 | 97.32 | 97.57 | 97.16 | 97.37 | 97.43 | 97.04 | 84.00 |
| **Imputation using propensity−score method** | | | | | | | | | | | |
| 10% | 99.17 | 99.07 | 99.13 | 98.88 | 98.90 | 99.04 | 99.03 | 99.06 | 99.00 | 98.84 | 93.08 |
| 20% | 98.76 | 98.72 | 98.73 | 98.65 | 98.57 | 98.66 | 98.52 | 98.68 | 98.53 | 98.25 | 90.95 |
| 30% | 98.06 | 98.22 | 98.16 | 98.18 | 98.16 | 97.93 | 97.94 | 97.99 | 97.70 | 97.63 | 87.38 |
| 40% | 97.48 | 97.48 | 97.04 | 97.28 | 97.10 | 96.90 | 96.87 | 96.72 | 96.75 | 96.27 | 81.39 |

Requested replicates per setting: 10000 | Imputations: 40 | Successful sample sizes: completion_summary.csv

# Table 3: Average interval length (expansion %) − Shrinkage

Scenario I: all means are 0 for T0,T1,...,T10

| | None adjustment | | | | | | | | | |
|---|---|---|---|---|---|---|---|---|---|---|
| **Missing rate** | **T1vs.T0** | **T2vs.T0** | **T3vs.T0** | **T4vs.T0** | **T5vs.T0** | **T6vs.T0** | **T7vs.T0** | **T8vs.T0** | **T9vs.T0** | **T10vs.T0** |
| 0% | 4.027 | 4.049 | 4.069 | 4.089 | 4.103 | 4.126 | 4.148 | 4.167 | 4.185 | 4.210 |
| **Imputation using linear regression** | | | | | | | | | | |
| 10% | 4.184 (3.90) | 4.201 (3.76) | 4.218 (3.66) | 4.239 (3.65) | 4.266 (3.97) | 4.284 (3.81) | 4.310 (3.90) | 4.324 (3.76) | 4.344 (3.80) | 4.364 (3.66) |
| 20% | 4.543 (12.80) | 4.561 (12.63) | 4.579 (12.55) | 4.604 (12.58) | 4.634 (12.93) | 4.657 (12.86) | 4.678 (12.78) | 4.699 (12.76) | 4.712 (12.59) | 4.747 (12.75) |
| 30% | 5.033 (24.96) | 5.079 (25.43) | 5.081 (24.89) | 5.114 (25.04) | 5.147 (25.44) | 5.170 (25.29) | 5.184 (24.97) | 5.222 (25.33) | 5.236 (25.12) | 5.254 (24.81) |
| 40% | 5.787 (43.70) | 5.799 (43.22) | 5.846 (43.67) | 5.887 (43.96) | 5.934 (44.61) | 5.961 (44.45) | 5.963 (43.76) | 6.008 (44.18) | 6.028 (44.04) | 6.072 (44.24) |
| **Imputation using predictive mean matching method** | | | | | | | | | | |
| 10% | 4.005 (−0.56) | 4.022 (−0.68) | 4.037 (−0.77) | 4.059 (−0.75) | 4.082 (−0.51) | 4.102 (−0.60) | 4.123 (−0.60) | 4.139 (−0.68) | 4.156 (−0.68) | 4.179 (−0.74) |
| 20% | 4.114 (2.16) | 4.130 (2.00) | 4.148 (1.94) | 4.172 (2.02) | 4.198 (2.30) | 4.217 (2.20) | 4.234 (2.09) | 4.257 (2.15) | 4.269 (2.01) | 4.302 (2.18) |
| 30% | 4.229 (5.00) | 4.260 (5.21) | 4.271 (4.96) | 4.291 (4.94) | 4.318 (5.24) | 4.337 (5.11) | 4.357 (5.05) | 4.386 (5.24) | 4.397 (5.08) | 4.417 (4.91) |
| 40% | 4.345 (7.89) | 4.364 (7.78) | 4.386 (7.80) | 4.419 (8.06) | 4.443 (8.29) | 4.468 (8.27) | 4.486 (8.16) | 4.504 (8.09) | 4.518 (7.96) | 4.545 (7.96) |
| **Imputation using propensity−score method** | | | | | | | | | | |
| 10% | 4.014 (−0.34) | 4.033 (−0.41) | 4.046 (−0.55) | 4.070 (−0.49) | 4.093 (−0.25) | 4.112 (−0.35) | 4.135 (−0.31) | 4.148 (−0.46) | 4.168 (−0.39) | 4.189 (−0.51) |
| 20% | 4.139 (2.78) | 4.157 (2.66) | 4.172 (2.55) | 4.200 (2.71) | 4.224 (2.95) | 4.242 (2.80) | 4.258 (2.65) | 4.282 (2.75) | 4.297 (2.69) | 4.325 (2.75) |
| 30% | 4.252 (5.59) | 4.282 (5.76) | 4.296 (5.58) | 4.313 (5.47) | 4.341 (5.81) | 4.362 (5.70) | 4.384 (5.68) | 4.409 (5.81) | 4.423 (5.70) | 4.436 (5.38) |
| 40% | 4.342 (7.82) | 4.361 (7.70) | 4.388 (7.85) | 4.418 (8.03) | 4.444 (8.30) | 4.465 (8.22) | 4.485 (8.13) | 4.502 (8.04) | 4.522 (8.06) | 4.544 (7.95) |

Requested replicates per setting: 10000 | Imputations: 40 | Successful sample sizes: completion_summary.csv

# Table 3: Average interval length (expansion %) − Shrinkage

Scenario I: all means are 0 for T0,T1,...,T10

| | Many−to−one adjustment | | | | | | | | | |
|---|---|---|---|---|---|---|---|---|---|---|
| **Missing rate** | **T1vs.T0** | **T2vs.T0** | **T3vs.T0** | **T4vs.T0** | **T5vs.T0** | **T6vs.T0** | **T7vs.T0** | **T8vs.T0** | **T9vs.T0** | **T10vs.T0** |
| 0% | 5.470 | 5.474 | 5.479 | 5.480 | 5.482 | 5.487 | 5.492 | 5.493 | 5.498 | 5.502 |
| **Imputation using linear regression** | | | | | | | | | | |
| 10% | 5.809 (6.20) | 5.815 (6.22) | 5.822 (6.26) | 5.829 (6.38) | 5.843 (6.59) | 5.849 (6.60) | 5.865 (6.79) | 5.873 (6.91) | 5.880 (6.95) | 5.889 (7.03) |
| 20% | 6.309 (15.33) | 6.319 (15.42) | 6.333 (15.59) | 6.347 (15.83) | 6.369 (16.18) | 6.388 (16.42) | 6.401 (16.56) | 6.417 (16.80) | 6.423 (16.82) | 6.449 (17.21) |
| 30% | 6.980 (27.59) | 7.029 (28.39) | 7.022 (28.17) | 7.054 (28.72) | 7.084 (29.21) | 7.104 (29.48) | 7.112 (29.51) | 7.151 (30.17) | 7.164 (30.30) | 7.178 (30.46) |
| 40% | 8.006 (46.36) | 8.016 (46.43) | 8.065 (47.20) | 8.110 (48.00) | 8.165 (48.93) | 8.192 (49.31) | 8.188 (49.09) | 8.240 (50.00) | 8.264 (50.30) | 8.315 (51.11) |
| **Imputation using predictive mean matching method** | | | | | | | | | | |
| 10% | 5.564 (1.70) | 5.569 (1.72) | 5.574 (1.74) | 5.579 (1.82) | 5.587 (1.92) | 5.594 (1.95) | 5.604 (2.04) | 5.611 (2.15) | 5.616 (2.14) | 5.625 (2.23) |
| 20% | 5.724 (4.63) | 5.728 (4.64) | 5.739 (4.75) | 5.749 (4.92) | 5.764 (5.14) | 5.776 (5.27) | 5.783 (5.30) | 5.797 (5.53) | 5.802 (5.52) | 5.822 (5.80) |
| 30% | 5.883 (7.54) | 5.908 (7.92) | 5.910 (7.87) | 5.926 (8.15) | 5.942 (8.39) | 5.958 (8.58) | 5.969 (8.69) | 5.991 (9.05) | 5.999 (9.11) | 6.011 (9.25) |
| 40% | 6.044 (10.49) | 6.060 (10.70) | 6.074 (10.86) | 6.104 (11.39) | 6.124 (11.71) | 6.145 (12.00) | 6.158 (12.13) | 6.173 (12.36) | 6.185 (12.50) | 6.207 (12.80) |
| **Imputation using propensity−score method** | | | | | | | | | | |
| 10% | 5.576 (1.94) | 5.583 (1.99) | 5.588 (1.99) | 5.595 (2.10) | 5.603 (2.20) | 5.610 (2.24) | 5.621 (2.36) | 5.626 (2.42) | 5.633 (2.46) | 5.641 (2.52) |
| 20% | 5.757 (5.25) | 5.765 (5.30) | 5.773 (5.38) | 5.788 (5.62) | 5.802 (5.84) | 5.812 (5.92) | 5.818 (5.94) | 5.834 (6.20) | 5.844 (6.28) | 5.858 (6.46) |
| 30% | 5.913 (8.09) | 5.938 (8.47) | 5.944 (8.50) | 5.958 (8.73) | 5.975 (9.00) | 5.992 (9.21) | 6.007 (9.38) | 6.025 (9.68) | 6.038 (9.82) | 6.044 (9.84) |
| 40% | 6.035 (10.32) | 6.051 (10.54) | 6.073 (10.85) | 6.100 (11.32) | 6.122 (11.68) | 6.141 (11.93) | 6.156 (12.10) | 6.170 (12.31) | 6.191 (12.60) | 6.208 (12.83) |

Requested replicates per setting: 10000 | Imputations: 40 | Successful sample sizes: completion_summary.csv

# Table 3: Average interval length (expansion %) − Shrinkage

Scenario II: mu0=0, mu1=1.5, mu2=2.0, ..., mu10=6.0

| | None adjustment | | | | | | | | | |
|---|---|---|---|---|---|---|---|---|---|---|
| **Missing rate** | **T1vs.T0** | **T2vs.T0** | **T3vs.T0** | **T4vs.T0** | **T5vs.T0** | **T6vs.T0** | **T7vs.T0** | **T8vs.T0** | **T9vs.T0** | **T10vs.T0** |
| 0% | 4.036 | 4.048 | 4.065 | 4.088 | 4.111 | 4.125 | 4.147 | 4.169 | 4.184 | 4.203 |
| **Imputation using linear regression** | | | | | | | | | | |
| 10% | 4.160 (3.07) | 4.180 (3.27) | 4.207 (3.50) | 4.223 (3.31) | 4.248 (3.34) | 4.261 (3.30) | 4.285 (3.31) | 4.300 (3.13) | 4.329 (3.46) | 4.347 (3.42) |
| 20% | 4.539 (12.47) | 4.562 (12.70) | 4.592 (12.98) | 4.606 (12.66) | 4.630 (12.63) | 4.662 (13.03) | 4.668 (12.56) | 4.697 (12.65) | 4.719 (12.78) | 4.747 (12.95) |
| 30% | 5.040 (24.88) | 5.072 (25.32) | 5.097 (25.39) | 5.121 (25.26) | 5.143 (25.10) | 5.155 (24.99) | 5.190 (25.15) | 5.218 (25.14) | 5.240 (25.25) | 5.267 (25.32) |
| 40% | 5.803 (43.80) | 5.846 (44.42) | 5.841 (43.71) | 5.872 (43.64) | 5.911 (43.78) | 5.931 (43.79) | 5.967 (43.88) | 6.016 (44.29) | 6.012 (43.69) | 6.095 (45.02) |
| **Imputation using predictive mean matching method** | | | | | | | | | | |
| 10% | 3.984 (−1.29) | 4.006 (−1.03) | 4.029 (−0.87) | 4.046 (−1.04) | 4.068 (−1.04) | 4.080 (−1.07) | 4.107 (−0.97) | 4.120 (−1.19) | 4.147 (−0.88) | 4.164 (−0.94) |
| 20% | 4.113 (1.92) | 4.128 (2.00) | 4.156 (2.26) | 4.168 (1.96) | 4.191 (1.95) | 4.218 (2.26) | 4.229 (1.98) | 4.244 (1.80) | 4.272 (2.11) | 4.298 (2.25) |
| 30% | 4.238 (5.02) | 4.266 (5.40) | 4.284 (5.39) | 4.310 (5.43) | 4.320 (5.08) | 4.336 (5.12) | 4.362 (5.18) | 4.387 (5.22) | 4.408 (5.35) | 4.426 (5.31) |
| 40% | 4.345 (7.65) | 4.374 (8.07) | 4.394 (8.10) | 4.409 (7.84) | 4.440 (8.00) | 4.453 (7.97) | 4.475 (7.91) | 4.484 (7.55) | 4.520 (8.03) | 4.544 (8.11) |
| **Imputation using propensity−score method** | | | | | | | | | | |
| 10% | 3.993 (−1.06) | 4.014 (−0.82) | 4.039 (−0.63) | 4.056 (−0.80) | 4.078 (−0.80) | 4.091 (−0.82) | 4.117 (−0.74) | 4.128 (−0.99) | 4.158 (−0.61) | 4.174 (−0.70) |
| 20% | 4.140 (2.59) | 4.153 (2.60) | 4.183 (2.91) | 4.195 (2.62) | 4.218 (2.60) | 4.245 (2.93) | 4.255 (2.59) | 4.276 (2.55) | 4.298 (2.73) | 4.328 (2.97) |
| 30% | 4.263 (5.63) | 4.293 (6.06) | 4.311 (6.06) | 4.339 (6.14) | 4.348 (5.76) | 4.368 (5.89) | 4.393 (5.93) | 4.422 (6.07) | 4.446 (6.26) | 4.471 (6.38) |
| 40% | 4.353 (7.87) | 4.378 (8.15) | 4.399 (8.23) | 4.419 (8.10) | 4.454 (8.35) | 4.481 (8.64) | 4.505 (8.62) | 4.520 (8.40) | 4.563 (9.06) | 4.593 (9.28) |

Requested replicates per setting: 10000 | Imputations: 40 | Successful sample sizes: completion_summary.csv

# Table 3: Average interval length (expansion %) − Shrinkage

Scenario II: mu0=0, mu1=1.5, mu2=2.0, ..., mu10=6.0

| | Many−to−one adjustment | | | | | | | | | |
|---|---|---|---|---|---|---|---|---|---|---|
| **Missing rate** | **T1vs.T0** | **T2vs.T0** | **T3vs.T0** | **T4vs.T0** | **T5vs.T0** | **T6vs.T0** | **T7vs.T0** | **T8vs.T0** | **T9vs.T0** | **T10vs.T0** |
| 0% | 5.474 | 5.476 | 5.478 | 5.482 | 5.485 | 5.489 | 5.493 | 5.496 | 5.499 | 5.503 |
| **Imputation using linear regression** | | | | | | | | | | |
| 10% | 5.791 (5.79) | 5.797 (5.87) | 5.811 (6.09) | 5.818 (6.13) | 5.829 (6.27) | 5.835 (6.30) | 5.843 (6.37) | 5.855 (6.53) | 5.867 (6.70) | 5.874 (6.74) |
| 20% | 6.300 (15.09) | 6.317 (15.36) | 6.341 (15.76) | 6.347 (15.78) | 6.365 (16.03) | 6.388 (16.39) | 6.390 (16.33) | 6.414 (16.70) | 6.426 (16.86) | 6.447 (17.16) |
| 30% | 6.988 (27.65) | 7.017 (28.14) | 7.039 (28.50) | 7.057 (28.74) | 7.083 (29.13) | 7.089 (29.15) | 7.122 (29.65) | 7.145 (30.00) | 7.166 (30.31) | 7.192 (30.69) |
| 40% | 8.023 (46.56) | 8.069 (47.35) | 8.056 (47.08) | 8.092 (47.61) | 8.131 (48.23) | 8.153 (48.53) | 8.192 (49.13) | 8.251 (50.12) | 8.236 (49.78) | 8.339 (51.55) |
| **Imputation using predictive mean matching method** | | | | | | | | | | |
| 10% | 5.549 (1.36) | 5.557 (1.48) | 5.565 (1.59) | 5.570 (1.61) | 5.578 (1.68) | 5.581 (1.67) | 5.590 (1.76) | 5.598 (1.85) | 5.607 (1.97) | 5.611 (1.96) |
| 20% | 5.717 (4.45) | 5.725 (4.55) | 5.741 (4.81) | 5.743 (4.77) | 5.759 (4.98) | 5.773 (5.17) | 5.778 (5.19) | 5.785 (5.26) | 5.800 (5.47) | 5.814 (5.65) |
| 30% | 5.893 (7.65) | 5.911 (7.94) | 5.922 (8.11) | 5.941 (8.37) | 5.950 (8.46) | 5.956 (8.52) | 5.974 (8.76) | 5.992 (9.02) | 6.007 (9.23) | 6.020 (9.39) |
| 40% | 6.040 (10.34) | 6.064 (10.74) | 6.080 (11.00) | 6.091 (11.12) | 6.116 (11.51) | 6.127 (11.62) | 6.145 (11.87) | 6.151 (11.91) | 6.183 (12.44) | 6.203 (12.71) |
| **Imputation using propensity−score method** | | | | | | | | | | |
| 10% | 5.562 (1.60) | 5.569 (1.70) | 5.578 (1.83) | 5.584 (1.87) | 5.592 (1.94) | 5.596 (1.96) | 5.604 (2.02) | 5.611 (2.10) | 5.624 (2.26) | 5.626 (2.24) |
| 20% | 5.754 (5.11) | 5.760 (5.18) | 5.779 (5.51) | 5.783 (5.49) | 5.797 (5.69) | 5.812 (5.89) | 5.814 (5.85) | 5.830 (6.08) | 5.839 (6.18) | 5.858 (6.46) |
| 30% | 5.929 (8.30) | 5.951 (8.68) | 5.962 (8.85) | 5.984 (9.16) | 5.992 (9.23) | 6.003 (9.36) | 6.020 (9.60) | 6.045 (10.00) | 6.062 (10.24) | 6.082 (10.51) |
| 40% | 6.057 (10.65) | 6.075 (10.95) | 6.095 (11.27) | 6.111 (11.48) | 6.143 (11.99) | 6.169 (12.40) | 6.188 (12.65) | 6.199 (12.79) | 6.241 (13.49) | 6.265 (13.85) |

Requested replicates per setting: 10000 | Imputations: 40 | Successful sample sizes: completion_summary.csv

# Table 2: Coverage (%) − Non−shrinkage

Scenario I: all means are 0 for T0,T1,...,T10

| | None adjustment | | | | | | | | | | |
|---|---|---|---|---|---|---|---|---|---|---|---|
| **Missing rate** | **T1vs.T0** | **T2vs.T0** | **T3vs.T0** | **T4vs.T0** | **T5vs.T0** | **T6vs.T0** | **T7vs.T0** | **T8vs.T0** | **T9vs.T0** | **T10vs.T0** | **Joint** |
| 0% | 95.05 | 94.71 | 94.69 | 95.12 | 94.77 | 95.24 | 94.91 | 94.98 | 94.89 | 95.16 | 72.79 |
| **Imputation using linear regression** | | | | | | | | | | | |
| 10% | 95.27 | 94.62 | 94.36 | 94.53 | 94.79 | 94.63 | 94.40 | 94.73 | 94.61 | 94.71 | 71.41 |
| 20% | 95.23 | 94.77 | 94.90 | 95.24 | 95.26 | 94.80 | 95.09 | 95.17 | 94.94 | 94.81 | 72.19 |
| 30% | 95.31 | 95.60 | 95.12 | 95.32 | 95.14 | 95.54 | 95.51 | 95.36 | 95.69 | 95.19 | 73.89 |
| 40% | 95.72 | 95.62 | 96.01 | 95.87 | 95.53 | 95.98 | 95.63 | 95.88 | 95.77 | 95.79 | 75.81 |
| **Imputation using predictive mean matching method** | | | | | | | | | | | |
| 10% | 94.28 | 93.58 | 93.46 | 93.61 | 93.90 | 93.64 | 93.38 | 93.78 | 93.84 | 93.75 | 67.40 |
| 20% | 92.97 | 92.76 | 92.43 | 92.60 | 93.02 | 92.45 | 92.62 | 92.91 | 92.65 | 92.68 | 63.02 |
| 30% | 91.14 | 91.47 | 91.02 | 91.35 | 91.06 | 91.62 | 91.12 | 91.22 | 91.72 | 91.11 | 57.60 |
| 40% | 89.64 | 89.36 | 89.86 | 89.79 | 89.60 | 89.59 | 89.76 | 89.60 | 89.64 | 89.41 | 51.42 |
| **Imputation using propensity−score method** | | | | | | | | | | | |
| 10% | 94.30 | 93.49 | 93.48 | 93.47 | 93.90 | 93.68 | 93.23 | 93.80 | 93.64 | 93.60 | 67.12 |
| 20% | 92.85 | 92.41 | 92.23 | 92.78 | 92.91 | 92.11 | 92.63 | 92.79 | 92.61 | 92.37 | 62.10 |
| 30% | 90.93 | 91.09 | 90.69 | 91.00 | 90.60 | 91.23 | 90.93 | 90.82 | 91.31 | 91.04 | 56.43 |
| 40% | 89.06 | 89.12 | 88.89 | 88.79 | 88.69 | 89.17 | 89.08 | 89.34 | 88.92 | 88.97 | 49.43 |

Requested replicates per setting: 10000 | Imputations: 40 | Successful sample sizes: completion_summary.csv

# Table 2: Coverage (%) − Non−shrinkage

Scenario I: all means are 0 for T0,T1,...,T10

| | Many−to−one adjustment | | | | | | | | | | |
|---|---|---|---|---|---|---|---|---|---|---|---|
| **Missing rate** | **T1vs.T0** | **T2vs.T0** | **T3vs.T0** | **T4vs.T0** | **T5vs.T0** | **T6vs.T0** | **T7vs.T0** | **T8vs.T0** | **T9vs.T0** | **T10vs.T0** | **Joint** |
| 0% | 98.96 | 98.91 | 99.15 | 99.18 | 98.96 | 99.02 | 99.03 | 99.04 | 98.90 | 99.17 | 93.36 |
| **Imputation using linear regression** | | | | | | | | | | | |
| 10% | 99.32 | 99.10 | 98.95 | 98.98 | 99.10 | 99.13 | 98.87 | 99.17 | 99.04 | 99.11 | 93.49 |
| 20% | 99.24 | 99.20 | 99.05 | 99.28 | 99.30 | 99.17 | 99.25 | 99.28 | 99.19 | 99.31 | 94.34 |
| 30% | 99.23 | 99.45 | 99.21 | 99.31 | 99.21 | 99.29 | 99.33 | 99.30 | 99.36 | 99.37 | 95.00 |
| 40% | 99.35 | 99.34 | 99.42 | 99.31 | 99.34 | 99.42 | 99.47 | 99.44 | 99.27 | 99.42 | 95.16 |
| **Imputation using predictive mean matching method** | | | | | | | | | | | |
| 10% | 99.01 | 98.74 | 98.70 | 98.75 | 98.83 | 98.82 | 98.55 | 98.86 | 98.73 | 98.81 | 91.87 |
| 20% | 98.55 | 98.43 | 98.28 | 98.54 | 98.53 | 98.29 | 98.56 | 98.63 | 98.38 | 98.38 | 89.88 |
| 30% | 97.85 | 97.97 | 97.64 | 97.81 | 97.63 | 98.00 | 98.04 | 98.01 | 98.07 | 98.02 | 86.78 |
| 40% | 97.20 | 97.01 | 97.38 | 97.06 | 96.91 | 97.12 | 97.09 | 97.34 | 96.81 | 97.16 | 82.11 |
| **Imputation using propensity−score method** | | | | | | | | | | | |
| 10% | 99.04 | 98.73 | 98.66 | 98.74 | 98.78 | 98.79 | 98.57 | 98.87 | 98.63 | 98.78 | 91.57 |
| 20% | 98.47 | 98.31 | 98.39 | 98.63 | 98.49 | 98.23 | 98.54 | 98.56 | 98.35 | 98.48 | 89.65 |
| 30% | 97.93 | 97.99 | 97.61 | 97.65 | 97.64 | 97.78 | 97.93 | 97.80 | 97.98 | 97.62 | 85.84 |
| 40% | 96.68 | 96.89 | 97.00 | 96.69 | 96.66 | 96.66 | 96.95 | 96.91 | 96.60 | 96.97 | 80.61 |

Requested replicates per setting: 10000 | Imputations: 40 | Successful sample sizes: completion_summary.csv

# Table 2: Coverage (%) − Non−shrinkage

Scenario II: mu0=0, mu1=1.5, mu2=2.0, ..., mu10=6.0

| | None adjustment | | | | | | | | | | |
|---|---|---|---|---|---|---|---|---|---|---|---|
| **Missing rate** | **T1vs.T0** | **T2vs.T0** | **T3vs.T0** | **T4vs.T0** | **T5vs.T0** | **T6vs.T0** | **T7vs.T0** | **T8vs.T0** | **T9vs.T0** | **T10vs.T0** | **Joint** |
| 0% | 94.98 | 95.11 | 95.25 | 95.36 | 95.05 | 94.76 | 95.06 | 95.46 | 94.70 | 95.03 | 73.01 |
| **Imputation using linear regression** | | | | | | | | | | | |
| 10% | 94.78 | 95.16 | 94.74 | 94.18 | 94.48 | 94.79 | 94.44 | 94.54 | 94.95 | 94.56 | 70.36 |
| 20% | 95.24 | 94.68 | 95.31 | 95.30 | 95.20 | 95.16 | 95.24 | 95.47 | 95.10 | 95.03 | 73.21 |
| 30% | 95.01 | 95.74 | 95.50 | 95.12 | 95.33 | 95.36 | 95.37 | 95.42 | 95.30 | 95.42 | 74.20 |
| 40% | 96.00 | 95.77 | 95.71 | 96.04 | 95.86 | 95.87 | 95.60 | 95.94 | 95.84 | 95.62 | 76.07 |
| **Imputation using predictive mean matching method** | | | | | | | | | | | |
| 10% | 93.85 | 94.00 | 93.77 | 93.07 | 93.28 | 93.67 | 93.56 | 93.73 | 93.84 | 93.48 | 66.07 |
| 20% | 93.28 | 92.64 | 92.99 | 92.98 | 93.04 | 92.97 | 92.93 | 93.02 | 92.85 | 92.82 | 63.72 |
| 30% | 91.49 | 91.77 | 91.43 | 91.62 | 91.43 | 91.30 | 91.73 | 91.71 | 91.45 | 91.48 | 58.77 |
| 40% | 89.38 | 89.24 | 88.86 | 89.35 | 89.15 | 89.25 | 89.41 | 89.58 | 89.51 | 89.31 | 50.35 |
| **Imputation using propensity−score method** | | | | | | | | | | | |
| 10% | 93.77 | 94.03 | 93.58 | 93.22 | 93.11 | 93.68 | 93.41 | 93.70 | 93.79 | 93.46 | 65.67 |
| 20% | 93.16 | 92.37 | 92.79 | 92.67 | 92.62 | 92.61 | 92.80 | 92.71 | 92.42 | 92.46 | 62.64 |
| 30% | 90.83 | 91.40 | 91.03 | 90.98 | 90.60 | 90.48 | 90.76 | 90.71 | 90.54 | 90.79 | 55.58 |
| 40% | 89.11 | 88.93 | 88.41 | 88.42 | 88.68 | 88.52 | 88.03 | 88.42 | 87.95 | 87.56 | 46.68 |

Requested replicates per setting: 10000 | Imputations: 40 | Successful sample sizes: completion_summary.csv

# Table 2: Coverage (%) − Non−shrinkage

Scenario II: mu0=0, mu1=1.5, mu2=2.0, ..., mu10=6.0

| | Many−to−one adjustment | | | | | | | | | | |
|---|---|---|---|---|---|---|---|---|---|---|---|
| **Missing rate** | **T1vs.T0** | **T2vs.T0** | **T3vs.T0** | **T4vs.T0** | **T5vs.T0** | **T6vs.T0** | **T7vs.T0** | **T8vs.T0** | **T9vs.T0** | **T10vs.T0** | **Joint** |
| 0% | 99.07 | 99.15 | 99.09 | 99.04 | 99.12 | 99.15 | 99.18 | 99.21 | 98.92 | 99.16 | 93.46 |
| **Imputation using linear regression** | | | | | | | | | | | |
| 10% | 99.19 | 99.17 | 99.18 | 99.10 | 99.16 | 99.16 | 99.15 | 99.17 | 99.23 | 98.99 | 93.84 |
| 20% | 99.26 | 99.18 | 99.26 | 99.10 | 99.18 | 99.18 | 99.23 | 99.34 | 99.19 | 98.88 | 94.41 |
| 30% | 99.21 | 99.30 | 99.28 | 99.31 | 99.22 | 99.26 | 99.19 | 99.18 | 99.24 | 99.22 | 94.59 |
| 40% | 99.30 | 99.40 | 99.28 | 99.35 | 99.29 | 99.32 | 99.38 | 99.32 | 99.40 | 99.33 | 95.41 |
| **Imputation using predictive mean matching method** | | | | | | | | | | | |
| 10% | 98.94 | 98.82 | 98.89 | 98.67 | 98.87 | 98.89 | 98.89 | 98.88 | 98.92 | 98.70 | 91.98 |
| 20% | 98.42 | 98.33 | 98.59 | 98.44 | 98.48 | 98.57 | 98.35 | 98.44 | 98.48 | 98.34 | 90.06 |
| 30% | 97.87 | 98.09 | 98.04 | 98.08 | 97.89 | 97.86 | 98.05 | 97.86 | 97.93 | 97.94 | 86.93 |
| 40% | 97.08 | 97.22 | 96.89 | 96.91 | 96.80 | 97.07 | 96.78 | 97.18 | 97.14 | 96.99 | 81.60 |
| **Imputation using propensity−score method** | | | | | | | | | | | |
| 10% | 98.94 | 98.82 | 98.90 | 98.70 | 98.82 | 98.84 | 98.92 | 98.91 | 98.84 | 98.63 | 91.83 |
| 20% | 98.36 | 98.29 | 98.60 | 98.30 | 98.38 | 98.43 | 98.30 | 98.48 | 98.31 | 98.29 | 89.43 |
| 30% | 97.47 | 97.79 | 97.72 | 97.82 | 97.76 | 97.44 | 97.69 | 97.70 | 97.43 | 97.43 | 85.26 |
| 40% | 96.82 | 97.00 | 96.56 | 96.81 | 96.53 | 96.65 | 96.37 | 96.23 | 96.46 | 96.14 | 78.96 |

Requested replicates per setting: 10000 | Imputations: 40 | Successful sample sizes: completion_summary.csv

# Table 3: Average interval length (expansion %) − Non−shrinkage

Scenario I: all means are 0 for T0,T1,...,T10

| | None adjustment | | | | | | | | | |
|---|---|---|---|---|---|---|---|---|---|---|
| **Missing rate** | **T1vs.T0** | **T2vs.T0** | **T3vs.T0** | **T4vs.T0** | **T5vs.T0** | **T6vs.T0** | **T7vs.T0** | **T8vs.T0** | **T9vs.T0** | **T10vs.T0** |
| 0% | 4.027 | 4.049 | 4.069 | 4.089 | 4.103 | 4.126 | 4.148 | 4.167 | 4.185 | 4.210 |
| **Imputation using linear regression** | | | | | | | | | | |
| 10% | 4.184 (3.90) | 4.201 (3.76) | 4.218 (3.66) | 4.239 (3.65) | 4.266 (3.97) | 4.284 (3.81) | 4.310 (3.90) | 4.324 (3.76) | 4.344 (3.80) | 4.364 (3.66) |
| 20% | 4.543 (12.80) | 4.561 (12.63) | 4.579 (12.55) | 4.604 (12.58) | 4.634 (12.93) | 4.657 (12.86) | 4.678 (12.78) | 4.699 (12.76) | 4.712 (12.59) | 4.747 (12.75) |
| 30% | 5.033 (24.96) | 5.079 (25.43) | 5.081 (24.89) | 5.114 (25.04) | 5.147 (25.44) | 5.170 (25.29) | 5.184 (24.97) | 5.222 (25.33) | 5.236 (25.12) | 5.254 (24.81) |
| 40% | 5.787 (43.70) | 5.799 (43.22) | 5.846 (43.67) | 5.887 (43.96) | 5.934 (44.61) | 5.961 (44.45) | 5.963 (43.76) | 6.008 (44.18) | 6.028 (44.04) | 6.072 (44.24) |
| **Imputation using predictive mean matching method** | | | | | | | | | | |
| 10% | 4.005 (−0.56) | 4.022 (−0.68) | 4.037 (−0.77) | 4.059 (−0.75) | 4.082 (−0.51) | 4.102 (−0.60) | 4.123 (−0.60) | 4.139 (−0.68) | 4.156 (−0.68) | 4.179 (−0.74) |
| 20% | 4.114 (2.16) | 4.130 (2.00) | 4.148 (1.94) | 4.172 (2.02) | 4.198 (2.30) | 4.217 (2.20) | 4.234 (2.09) | 4.257 (2.15) | 4.269 (2.01) | 4.302 (2.18) |
| 30% | 4.229 (5.00) | 4.260 (5.21) | 4.271 (4.96) | 4.291 (4.94) | 4.318 (5.24) | 4.337 (5.11) | 4.357 (5.05) | 4.386 (5.24) | 4.397 (5.08) | 4.417 (4.91) |
| 40% | 4.345 (7.89) | 4.364 (7.78) | 4.386 (7.80) | 4.419 (8.06) | 4.443 (8.29) | 4.468 (8.27) | 4.486 (8.16) | 4.504 (8.09) | 4.518 (7.96) | 4.545 (7.96) |
| **Imputation using propensity−score method** | | | | | | | | | | |
| 10% | 4.014 (−0.34) | 4.033 (−0.41) | 4.046 (−0.55) | 4.070 (−0.49) | 4.093 (−0.25) | 4.112 (−0.35) | 4.135 (−0.31) | 4.148 (−0.46) | 4.168 (−0.39) | 4.189 (−0.51) |
| 20% | 4.139 (2.78) | 4.157 (2.66) | 4.172 (2.55) | 4.200 (2.71) | 4.224 (2.95) | 4.242 (2.80) | 4.258 (2.65) | 4.282 (2.75) | 4.297 (2.69) | 4.325 (2.75) |
| 30% | 4.252 (5.59) | 4.282 (5.76) | 4.296 (5.58) | 4.313 (5.47) | 4.341 (5.81) | 4.362 (5.70) | 4.384 (5.68) | 4.409 (5.81) | 4.423 (5.70) | 4.436 (5.38) |
| 40% | 4.342 (7.82) | 4.361 (7.70) | 4.388 (7.85) | 4.418 (8.03) | 4.444 (8.30) | 4.465 (8.22) | 4.485 (8.13) | 4.502 (8.04) | 4.522 (8.06) | 4.544 (7.95) |

Requested replicates per setting: 10000 | Imputations: 40 | Successful sample sizes: completion_summary.csv

# Table 3: Average interval length (expansion %) − Non−shrinkage

Scenario I: all means are 0 for T0,T1,...,T10

| | Many−to−one adjustment | | | | | | | | | |
|---|---|---|---|---|---|---|---|---|---|---|
| **Missing rate** | **T1vs.T0** | **T2vs.T0** | **T3vs.T0** | **T4vs.T0** | **T5vs.T0** | **T6vs.T0** | **T7vs.T0** | **T8vs.T0** | **T9vs.T0** | **T10vs.T0** |
| 0% | 5.405 | 5.435 | 5.461 | 5.489 | 5.508 | 5.539 | 5.569 | 5.594 | 5.618 | 5.652 |
| **Imputation using linear regression** | | | | | | | | | | |
| 10% | 5.756 (6.49) | 5.779 (6.34) | 5.802 (6.24) | 5.831 (6.22) | 5.868 (6.54) | 5.893 (6.38) | 5.929 (6.47) | 5.948 (6.32) | 5.976 (6.37) | 6.003 (6.21) |
| 20% | 6.248 (15.60) | 6.273 (15.42) | 6.298 (15.33) | 6.332 (15.35) | 6.374 (15.71) | 6.406 (15.64) | 6.434 (15.55) | 6.463 (15.53) | 6.481 (15.37) | 6.530 (15.53) |
| 30% | 6.918 (27.99) | 6.981 (28.46) | 6.985 (27.91) | 7.030 (28.06) | 7.075 (28.45) | 7.107 (28.31) | 7.127 (27.99) | 7.180 (28.34) | 7.198 (28.13) | 7.224 (27.81) |
| 40% | 7.942 (46.94) | 7.959 (46.44) | 8.023 (46.91) | 8.080 (47.18) | 8.143 (47.84) | 8.181 (47.69) | 8.185 (46.98) | 8.247 (47.42) | 8.275 (47.29) | 8.337 (47.50) |
| **Imputation using predictive mean matching method** | | | | | | | | | | |
| 10% | 5.510 (1.95) | 5.534 (1.82) | 5.555 (1.72) | 5.585 (1.74) | 5.617 (1.98) | 5.644 (1.89) | 5.674 (1.88) | 5.695 (1.79) | 5.719 (1.80) | 5.750 (1.73) |
| 20% | 5.661 (4.74) | 5.684 (4.58) | 5.707 (4.51) | 5.741 (4.59) | 5.776 (4.87) | 5.803 (4.76) | 5.827 (4.64) | 5.858 (4.71) | 5.875 (4.56) | 5.920 (4.74) |
| 30% | 5.817 (7.63) | 5.861 (7.84) | 5.876 (7.59) | 5.904 (7.56) | 5.941 (7.86) | 5.968 (7.73) | 5.996 (7.67) | 6.034 (7.86) | 6.050 (7.69) | 6.077 (7.52) |
| 40% | 5.976 (10.57) | 6.003 (10.45) | 6.033 (10.47) | 6.079 (10.73) | 6.111 (10.95) | 6.145 (10.93) | 6.171 (10.82) | 6.196 (10.75) | 6.215 (10.62) | 6.252 (10.61) |
| **Imputation using propensity−score method** | | | | | | | | | | |
| 10% | 5.522 (2.16) | 5.548 (2.08) | 5.567 (1.94) | 5.599 (1.99) | 5.631 (2.23) | 5.658 (2.13) | 5.689 (2.17) | 5.707 (2.01) | 5.735 (2.08) | 5.763 (1.96) |
| 20% | 5.693 (5.33) | 5.718 (5.20) | 5.739 (5.08) | 5.778 (5.25) | 5.810 (5.49) | 5.835 (5.33) | 5.857 (5.17) | 5.890 (5.28) | 5.911 (5.21) | 5.950 (5.27) |
| 30% | 5.846 (8.17) | 5.888 (8.34) | 5.907 (8.16) | 5.931 (8.04) | 5.970 (8.38) | 5.998 (8.27) | 6.028 (8.25) | 6.063 (8.37) | 6.082 (8.26) | 6.101 (7.94) |
| 40% | 5.966 (10.38) | 5.992 (10.25) | 6.029 (10.40) | 6.070 (10.58) | 6.106 (10.86) | 6.136 (10.77) | 6.163 (10.67) | 6.187 (10.58) | 6.214 (10.60) | 6.245 (10.49) |

Requested replicates per setting: 10000 | Imputations: 40 | Successful sample sizes: completion_summary.csv

# Table 3: Average interval length (expansion %) − Non−shrinkage

Scenario II: mu0=0, mu1=1.5, mu2=2.0, ..., mu10=6.0

| | None adjustment | | | | | | | | | |
|---|---|---|---|---|---|---|---|---|---|---|
| **Missing rate** | **T1vs.T0** | **T2vs.T0** | **T3vs.T0** | **T4vs.T0** | **T5vs.T0** | **T6vs.T0** | **T7vs.T0** | **T8vs.T0** | **T9vs.T0** | **T10vs.T0** |
| 0% | 4.036 | 4.048 | 4.065 | 4.088 | 4.111 | 4.125 | 4.147 | 4.169 | 4.184 | 4.203 |
| **Imputation using linear regression** | | | | | | | | | | |
| 10% | 4.160 (3.07) | 4.180 (3.27) | 4.207 (3.50) | 4.223 (3.31) | 4.248 (3.34) | 4.261 (3.30) | 4.285 (3.31) | 4.300 (3.13) | 4.329 (3.46) | 4.347 (3.42) |
| 20% | 4.539 (12.47) | 4.562 (12.70) | 4.592 (12.98) | 4.606 (12.66) | 4.630 (12.63) | 4.662 (13.03) | 4.668 (12.56) | 4.697 (12.65) | 4.719 (12.78) | 4.747 (12.95) |
| 30% | 5.040 (24.88) | 5.072 (25.32) | 5.097 (25.39) | 5.121 (25.26) | 5.143 (25.10) | 5.155 (24.99) | 5.190 (25.15) | 5.218 (25.14) | 5.240 (25.25) | 5.267 (25.32) |
| 40% | 5.803 (43.80) | 5.846 (44.42) | 5.841 (43.71) | 5.872 (43.64) | 5.911 (43.78) | 5.931 (43.79) | 5.967 (43.88) | 6.016 (44.29) | 6.012 (43.69) | 6.095 (45.02) |
| **Imputation using predictive mean matching method** | | | | | | | | | | |
| 10% | 3.984 (−1.29) | 4.006 (−1.03) | 4.029 (−0.87) | 4.046 (−1.04) | 4.068 (−1.04) | 4.080 (−1.07) | 4.107 (−0.97) | 4.120 (−1.19) | 4.147 (−0.88) | 4.164 (−0.94) |
| 20% | 4.113 (1.92) | 4.128 (2.00) | 4.156 (2.26) | 4.168 (1.96) | 4.191 (1.95) | 4.218 (2.26) | 4.229 (1.98) | 4.244 (1.80) | 4.272 (2.11) | 4.298 (2.25) |
| 30% | 4.238 (5.02) | 4.266 (5.40) | 4.284 (5.39) | 4.310 (5.43) | 4.320 (5.08) | 4.336 (5.12) | 4.362 (5.18) | 4.387 (5.22) | 4.408 (5.35) | 4.426 (5.31) |
| 40% | 4.345 (7.65) | 4.374 (8.07) | 4.394 (8.10) | 4.409 (7.84) | 4.440 (8.00) | 4.453 (7.97) | 4.475 (7.91) | 4.484 (7.55) | 4.520 (8.03) | 4.544 (8.11) |
| **Imputation using propensity−score method** | | | | | | | | | | |
| 10% | 3.993 (−1.06) | 4.014 (−0.82) | 4.039 (−0.63) | 4.056 (−0.80) | 4.078 (−0.80) | 4.091 (−0.82) | 4.117 (−0.74) | 4.128 (−0.99) | 4.158 (−0.61) | 4.174 (−0.70) |
| 20% | 4.140 (2.59) | 4.153 (2.60) | 4.183 (2.91) | 4.195 (2.62) | 4.218 (2.60) | 4.245 (2.93) | 4.255 (2.59) | 4.276 (2.55) | 4.298 (2.73) | 4.328 (2.97) |
| 30% | 4.263 (5.63) | 4.293 (6.06) | 4.311 (6.06) | 4.339 (6.14) | 4.348 (5.76) | 4.368 (5.89) | 4.393 (5.93) | 4.422 (6.07) | 4.446 (6.26) | 4.471 (6.38) |
| 40% | 4.353 (7.87) | 4.378 (8.15) | 4.399 (8.23) | 4.419 (8.10) | 4.454 (8.35) | 4.481 (8.64) | 4.505 (8.62) | 4.520 (8.40) | 4.563 (9.06) | 4.593 (9.28) |

Requested replicates per setting: 10000 | Imputations: 40 | Successful sample sizes: completion_summary.csv

# Table 3: Average interval length (expansion %) − Non−shrinkage

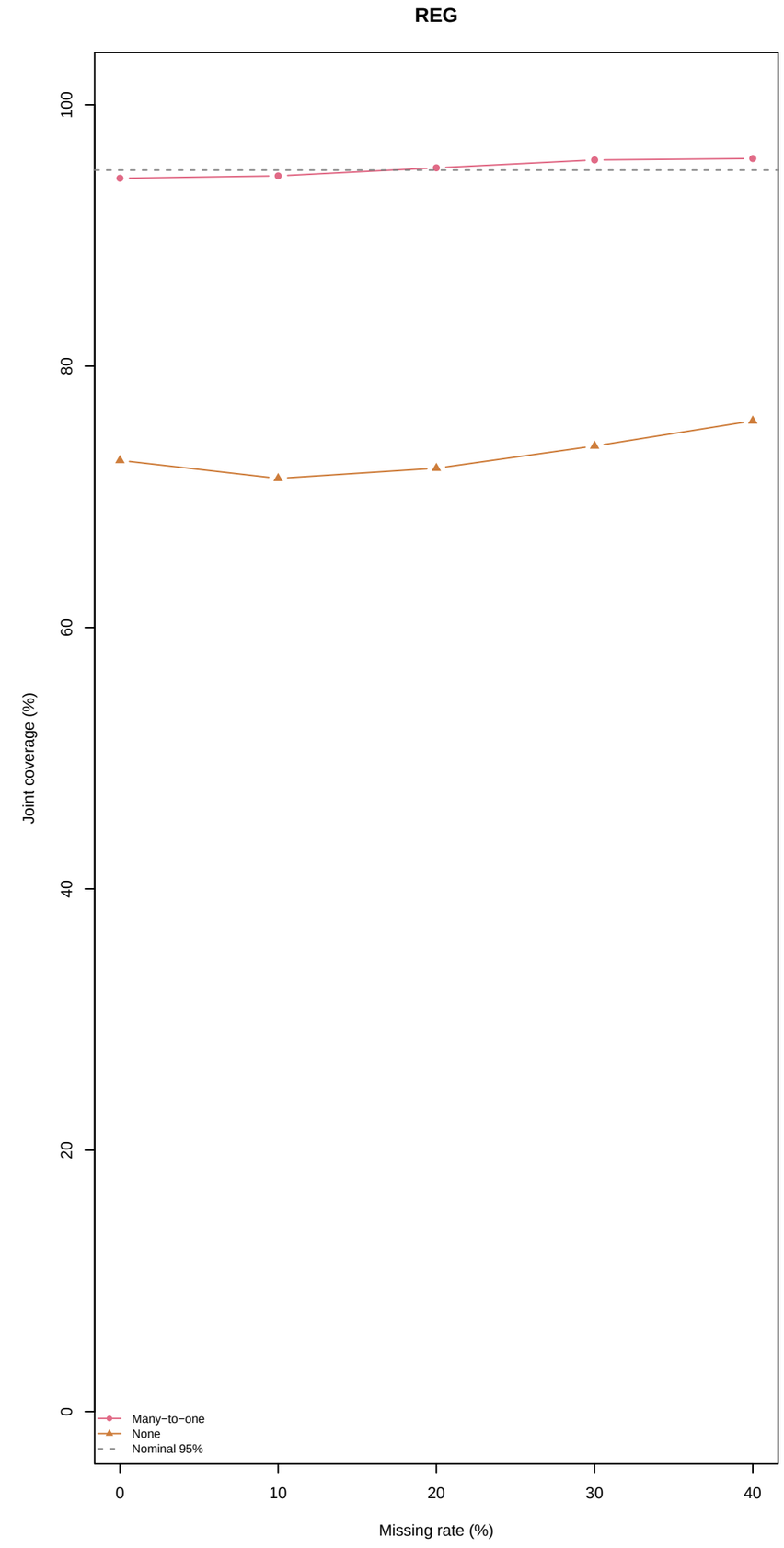
Scenario II: mu0=0, mu1=1.5, mu2=2.0, ..., mu10=6.0

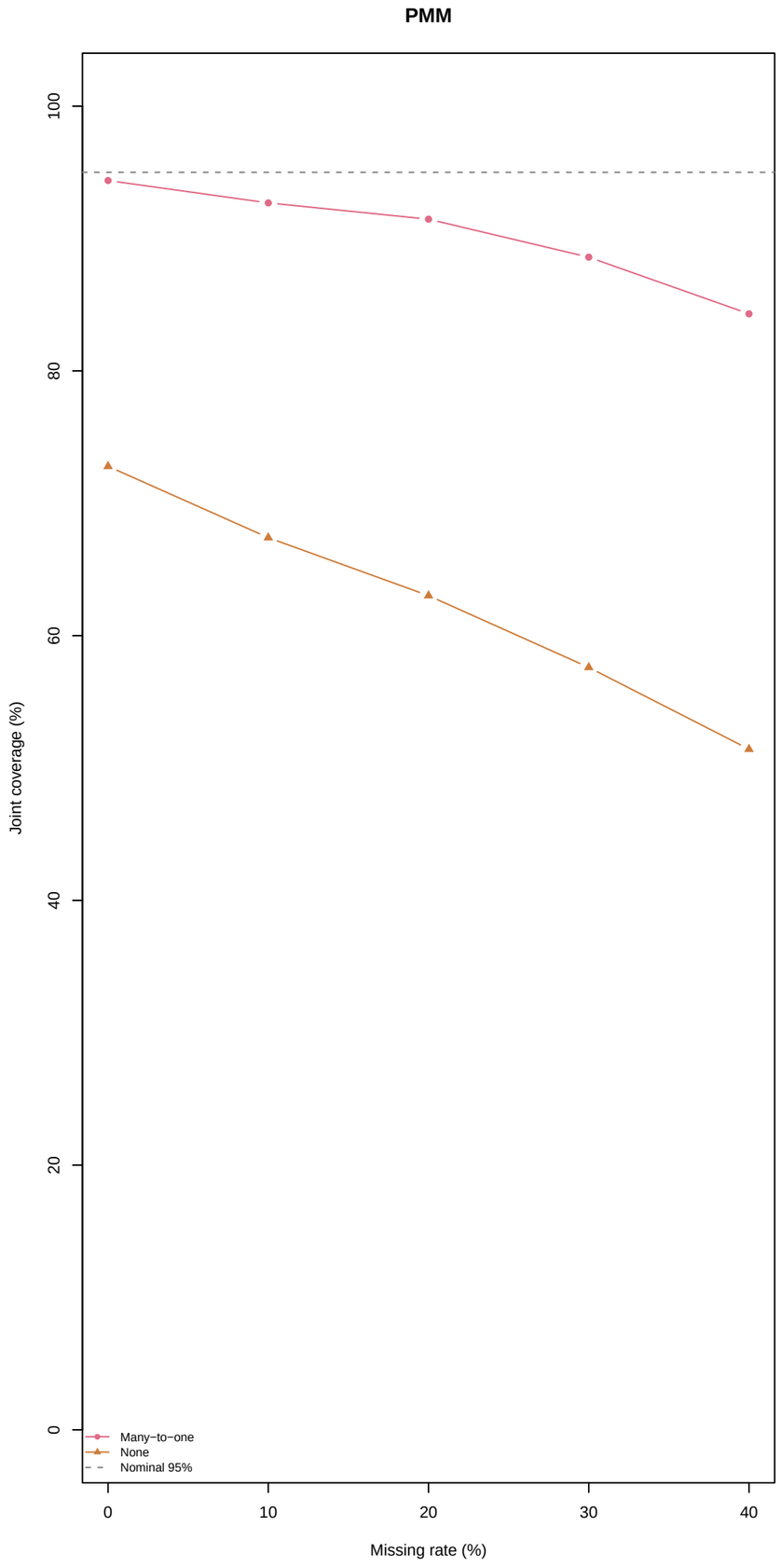
| | Many−to−one adjustment | | | | | | | | | |
|---|---|---|---|---|---|---|---|---|---|---|
| **Missing rate** | **T1vs.T0** | **T2vs.T0** | **T3vs.T0** | **T4vs.T0** | **T5vs.T0** | **T6vs.T0** | **T7vs.T0** | **T8vs.T0** | **T9vs.T0** | **T10vs.T0** |
| 0% | 5.417 | 5.433 | 5.457 | 5.488 | 5.519 | 5.538 | 5.568 | 5.598 | 5.618 | 5.643 |
| **Imputation using linear regression** | | | | | | | | | | |
| 10% | 5.725 (5.69) | 5.753 (5.88) | 5.790 (6.11) | 5.813 (5.91) | 5.847 (5.94) | 5.865 (5.90) | 5.898 (5.92) | 5.919 (5.72) | 5.958 (6.06) | 5.983 (6.02) |
| 20% | 6.242 (15.24) | 6.273 (15.46) | 6.316 (15.74) | 6.334 (15.41) | 6.368 (15.38) | 6.412 (15.78) | 6.421 (15.31) | 6.460 (15.39) | 6.490 (15.53) | 6.530 (15.70) |
| 30% | 6.929 (27.90) | 6.973 (28.34) | 7.007 (28.42) | 7.040 (28.28) | 7.070 (28.10) | 7.087 (27.98) | 7.136 (28.15) | 7.174 (28.14) | 7.205 (28.26) | 7.242 (28.32) |
| 40% | 7.961 (46.96) | 8.020 (47.61) | 8.014 (46.87) | 8.057 (46.80) | 8.110 (46.95) | 8.137 (46.94) | 8.188 (47.04) | 8.256 (47.47) | 8.250 (46.85) | 8.365 (48.23) |
| **Imputation using predictive mean matching method** | | | | | | | | | | |
| 10% | 5.484 (1.23) | 5.514 (1.49) | 5.546 (1.65) | 5.569 (1.47) | 5.600 (1.47) | 5.617 (1.43) | 5.654 (1.53) | 5.671 (1.30) | 5.709 (1.63) | 5.732 (1.57) |
| 20% | 5.660 (4.48) | 5.680 (4.55) | 5.719 (4.81) | 5.735 (4.50) | 5.767 (4.49) | 5.804 (4.80) | 5.820 (4.51) | 5.840 (4.32) | 5.879 (4.64) | 5.914 (4.79) |
| 30% | 5.830 (7.62) | 5.868 (8.01) | 5.893 (7.99) | 5.929 (8.03) | 5.942 (7.67) | 5.964 (7.70) | 6.000 (7.76) | 6.035 (7.80) | 6.064 (7.94) | 6.089 (7.90) |
| 40% | 5.975 (10.30) | 6.016 (10.73) | 6.044 (10.76) | 6.064 (10.48) | 6.107 (10.65) | 6.126 (10.61) | 6.156 (10.55) | 6.168 (10.18) | 6.217 (10.67) | 6.250 (10.76) |
| **Imputation using propensity−score method** | | | | | | | | | | |
| 10% | 5.496 (1.45) | 5.525 (1.70) | 5.559 (1.88) | 5.582 (1.71) | 5.613 (1.71) | 5.630 (1.67) | 5.666 (1.76) | 5.682 (1.50) | 5.724 (1.89) | 5.745 (1.80) |
| 20% | 5.694 (5.12) | 5.712 (5.12) | 5.753 (5.43) | 5.771 (5.14) | 5.801 (5.11) | 5.840 (5.45) | 5.852 (5.10) | 5.881 (5.05) | 5.912 (5.24) | 5.953 (5.49) |
| 30% | 5.861 (8.19) | 5.902 (8.62) | 5.927 (8.62) | 5.966 (8.70) | 5.977 (8.30) | 6.005 (8.43) | 6.041 (8.49) | 6.080 (8.61) | 6.113 (8.81) | 6.147 (8.93) |
| 40% | 5.984 (10.47) | 6.018 (10.76) | 6.048 (10.83) | 6.075 (10.70) | 6.123 (10.95) | 6.160 (11.23) | 6.193 (11.22) | 6.214 (11.00) | 6.274 (11.67) | 6.315 (11.89) |

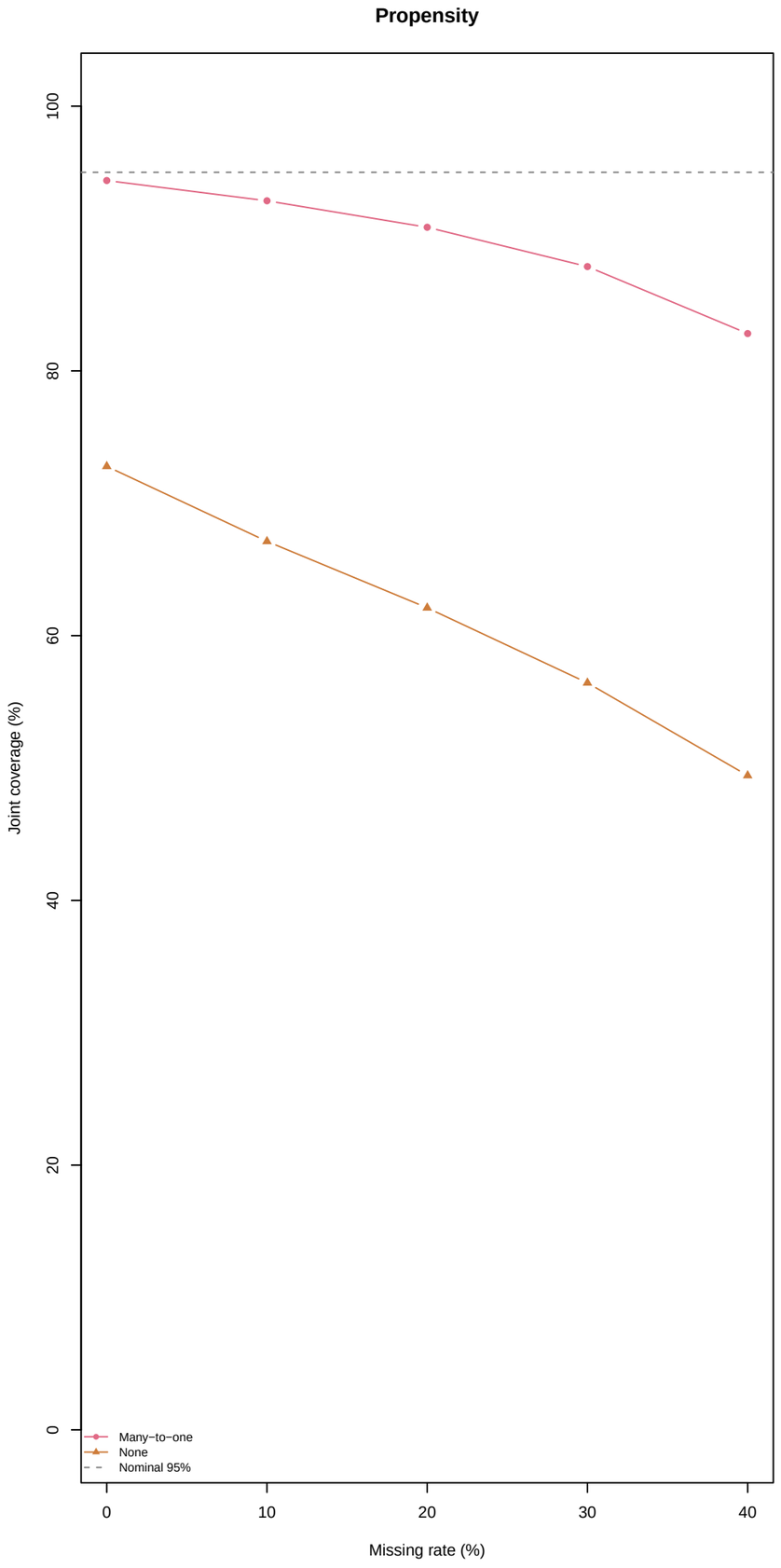
Requested replicates per setting: 10000 | Imputations: 40 | Successful sample sizes: completion_summary.csv

# Joint coverage: Shrinkage

Scenario I: all means are 0 for T0,T1,...,T10

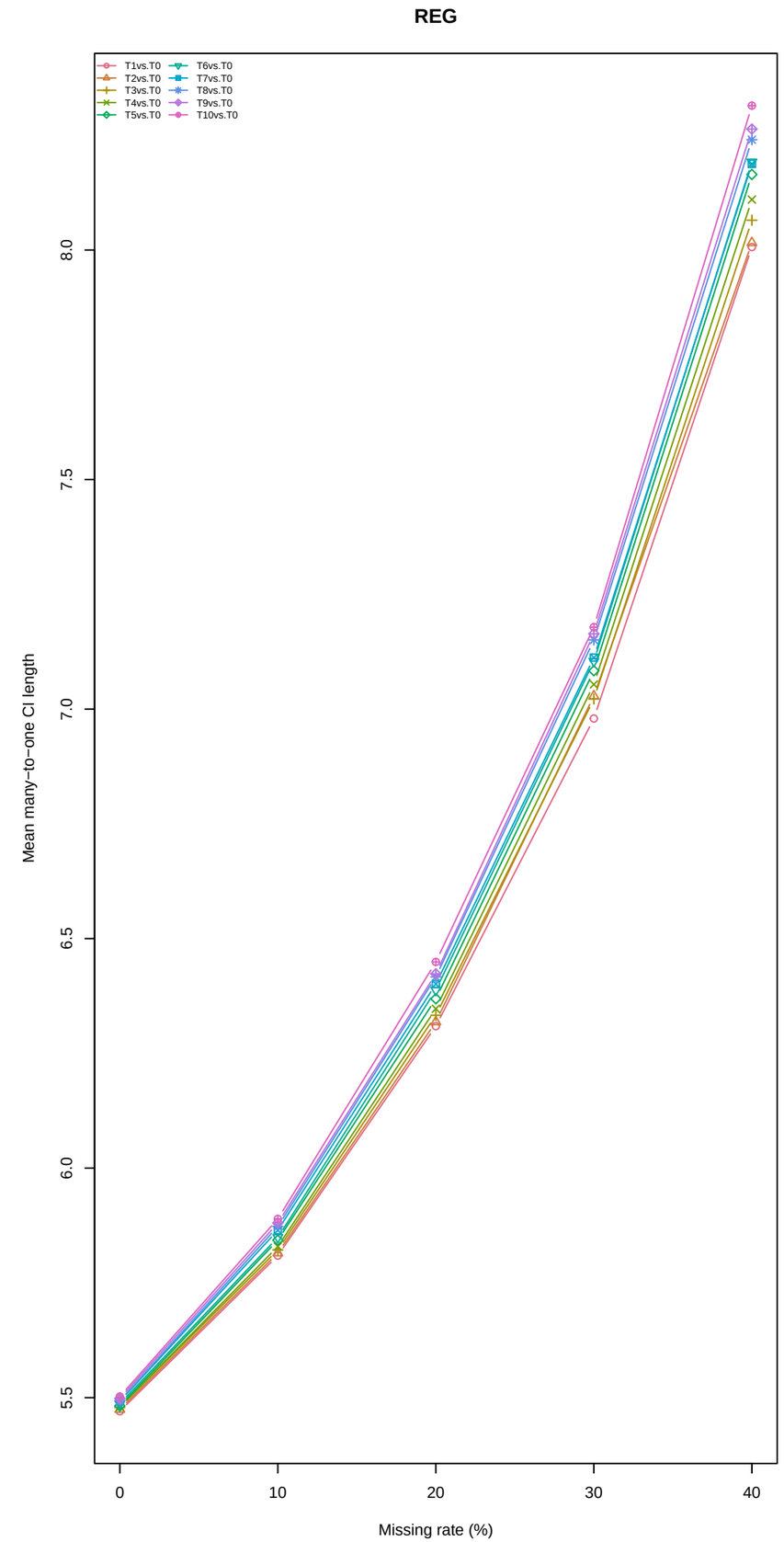


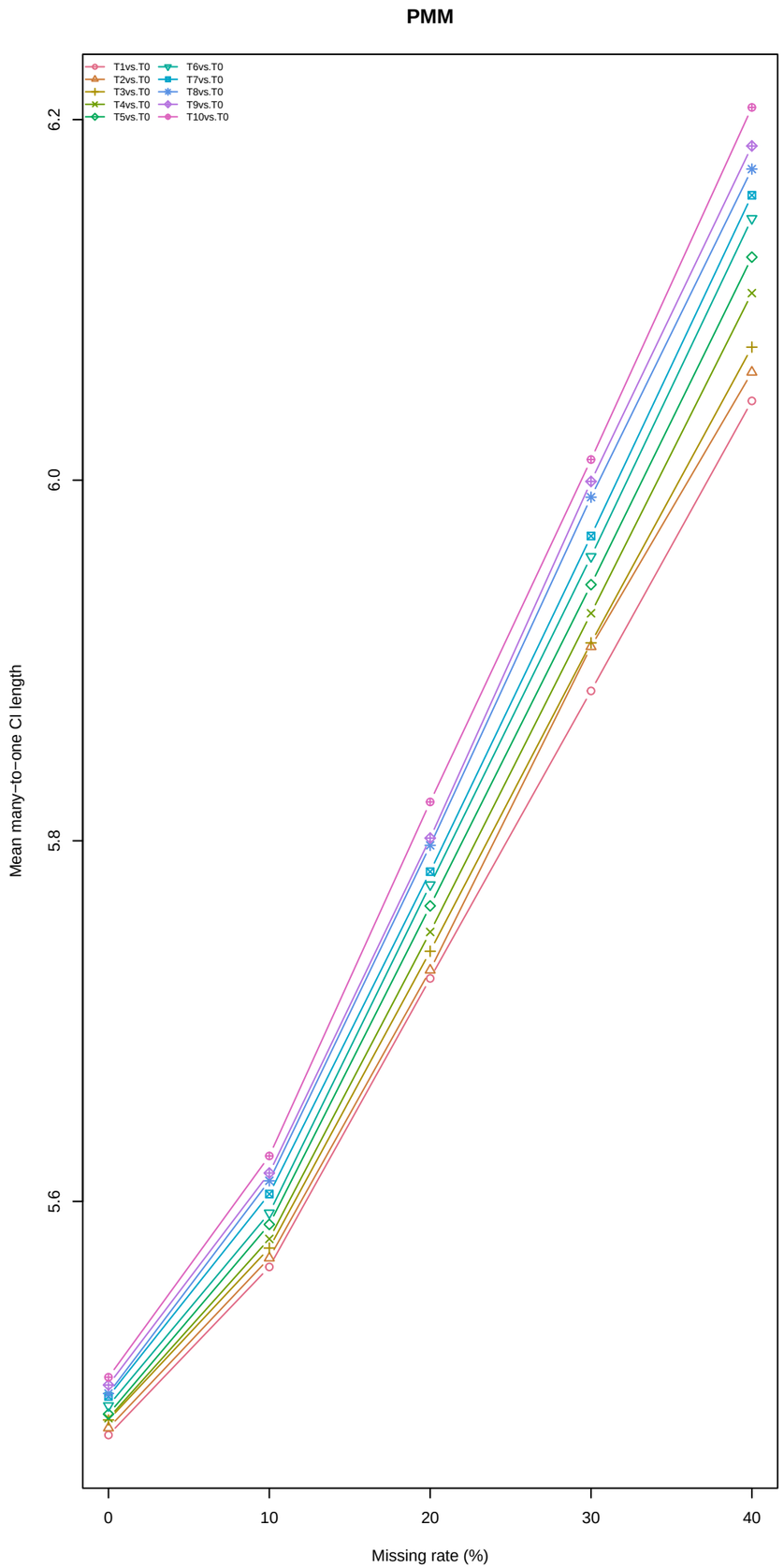



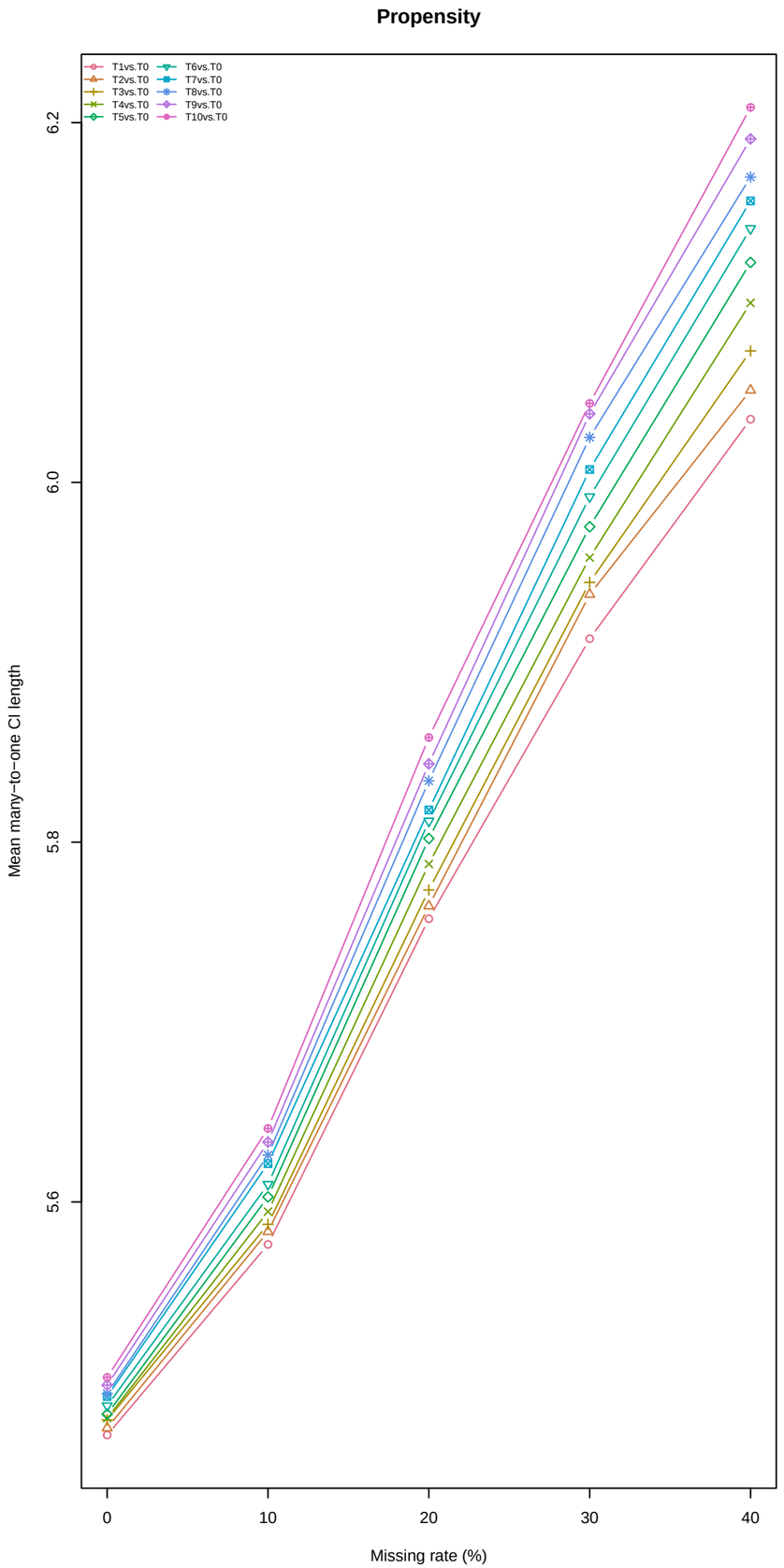



# Interval lengths: Shrinkage

Scenario I: all means are 0 for T0,T1,...,T10

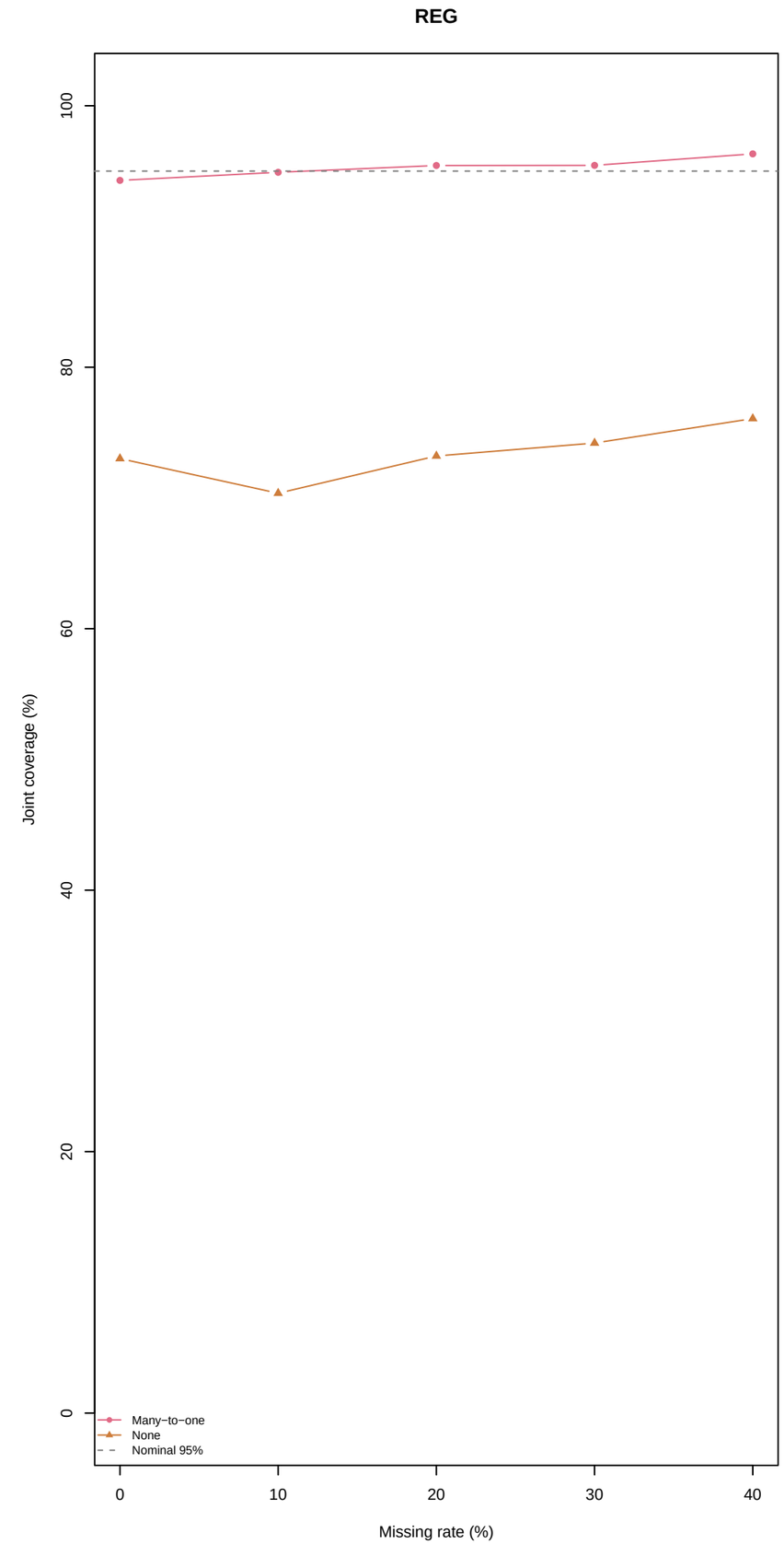


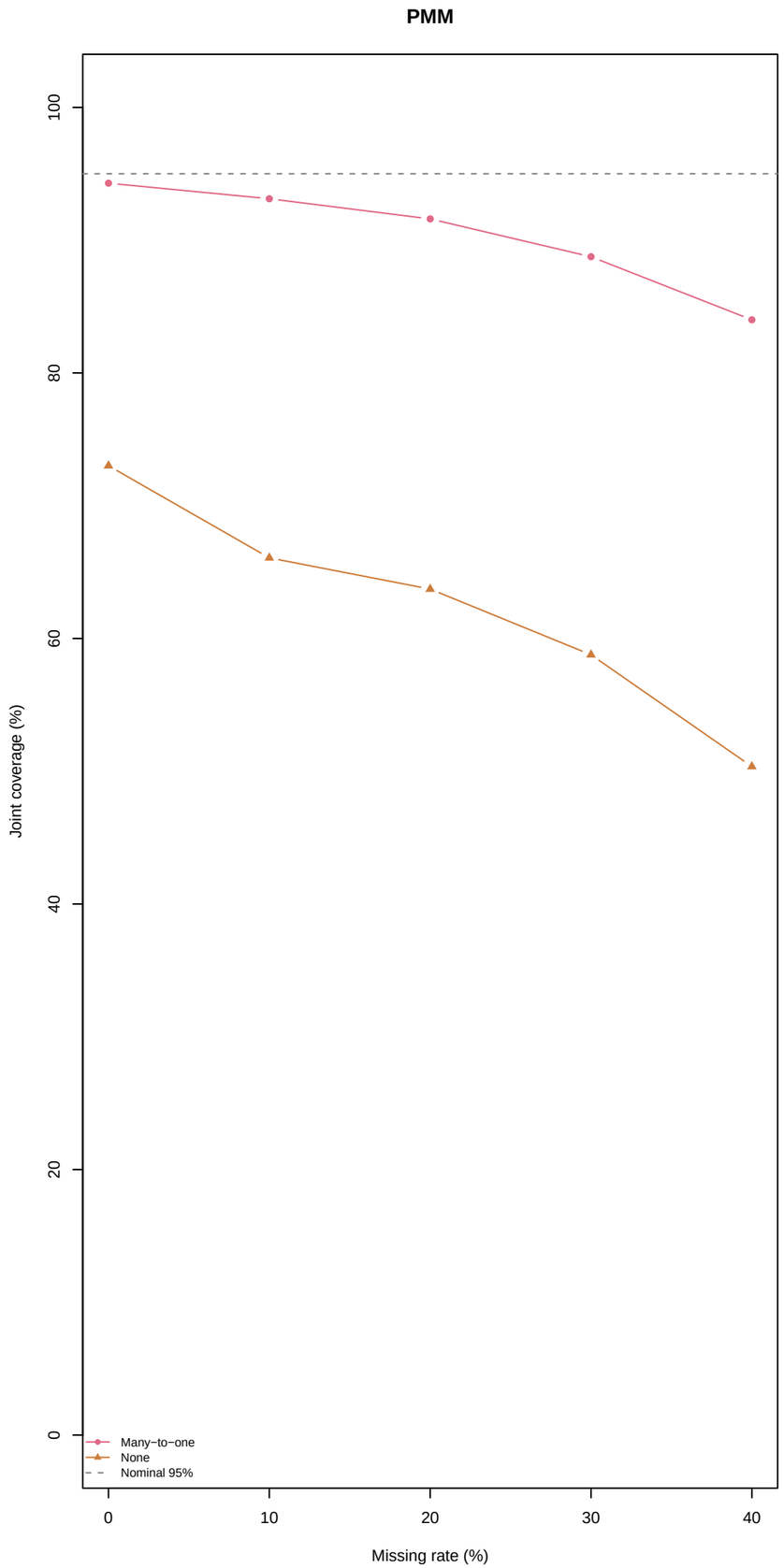



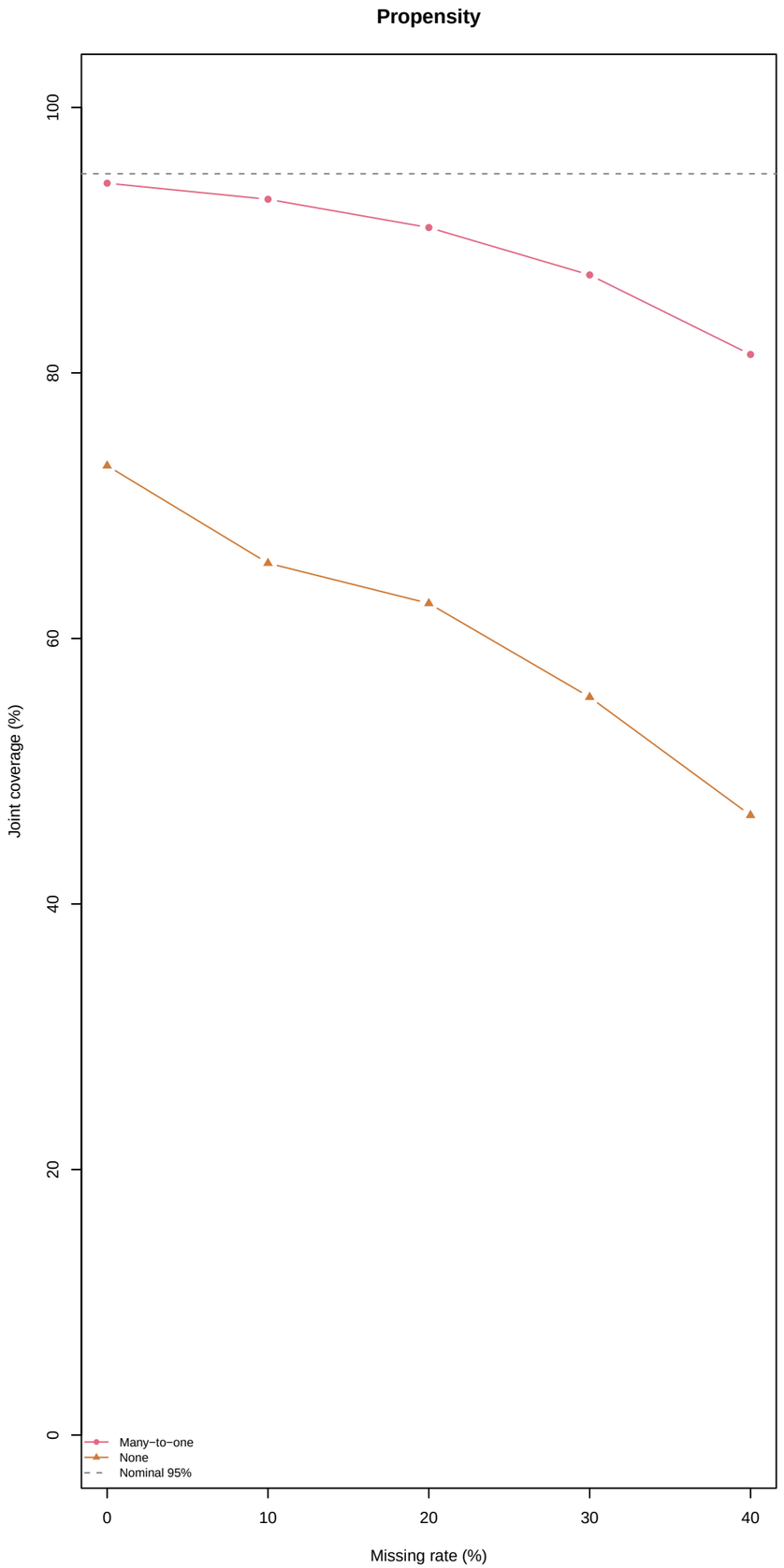



# Joint coverage: Shrinkage

Scenario II: mu0=0, mu1=1.5, mu2=2.0, ..., mu10=6.0

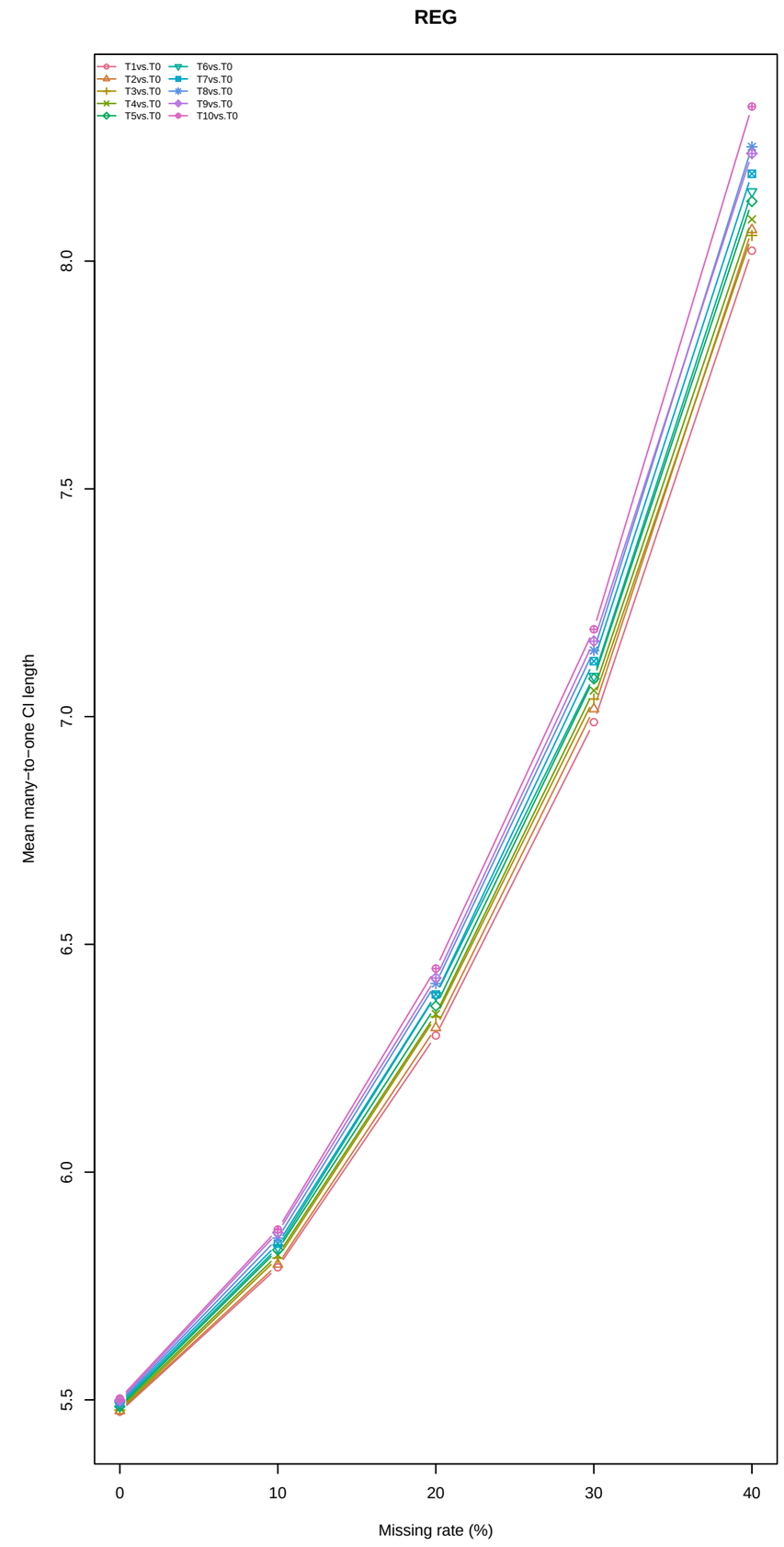


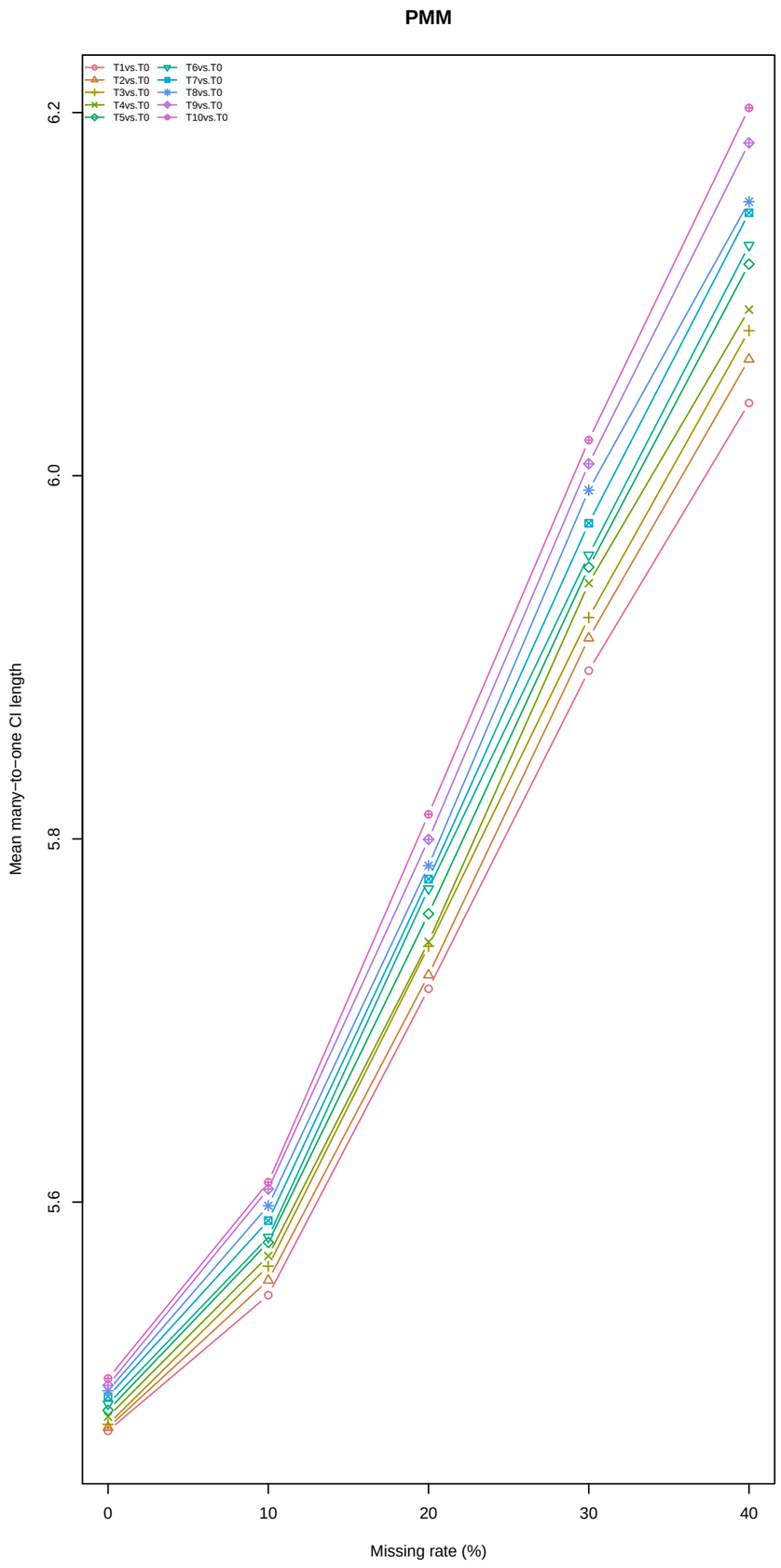



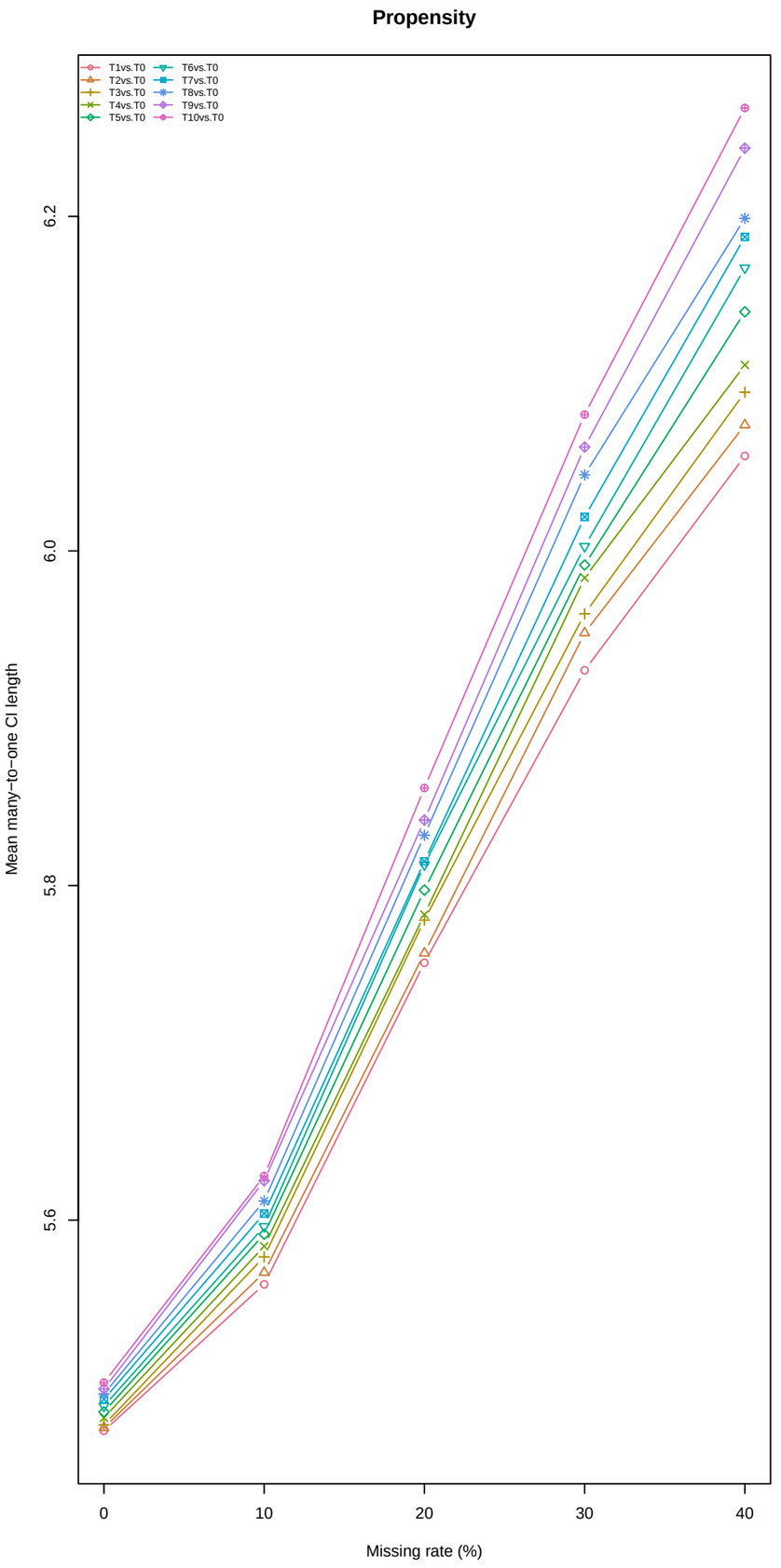



# Interval lengths: Shrinkage

Scenario II: mu0=0, mu1=1.5, mu2=2.0, ..., mu10=6.0

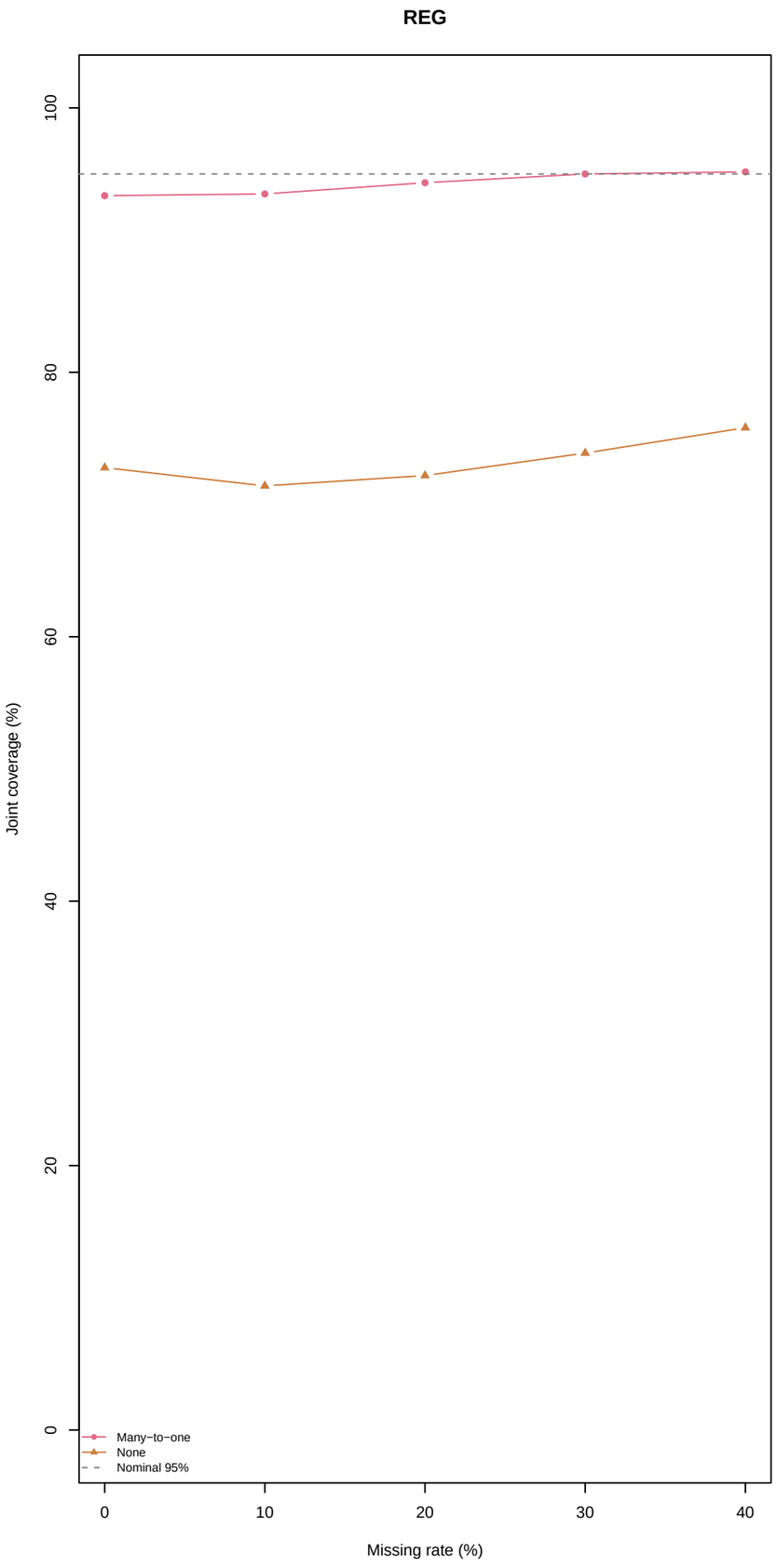


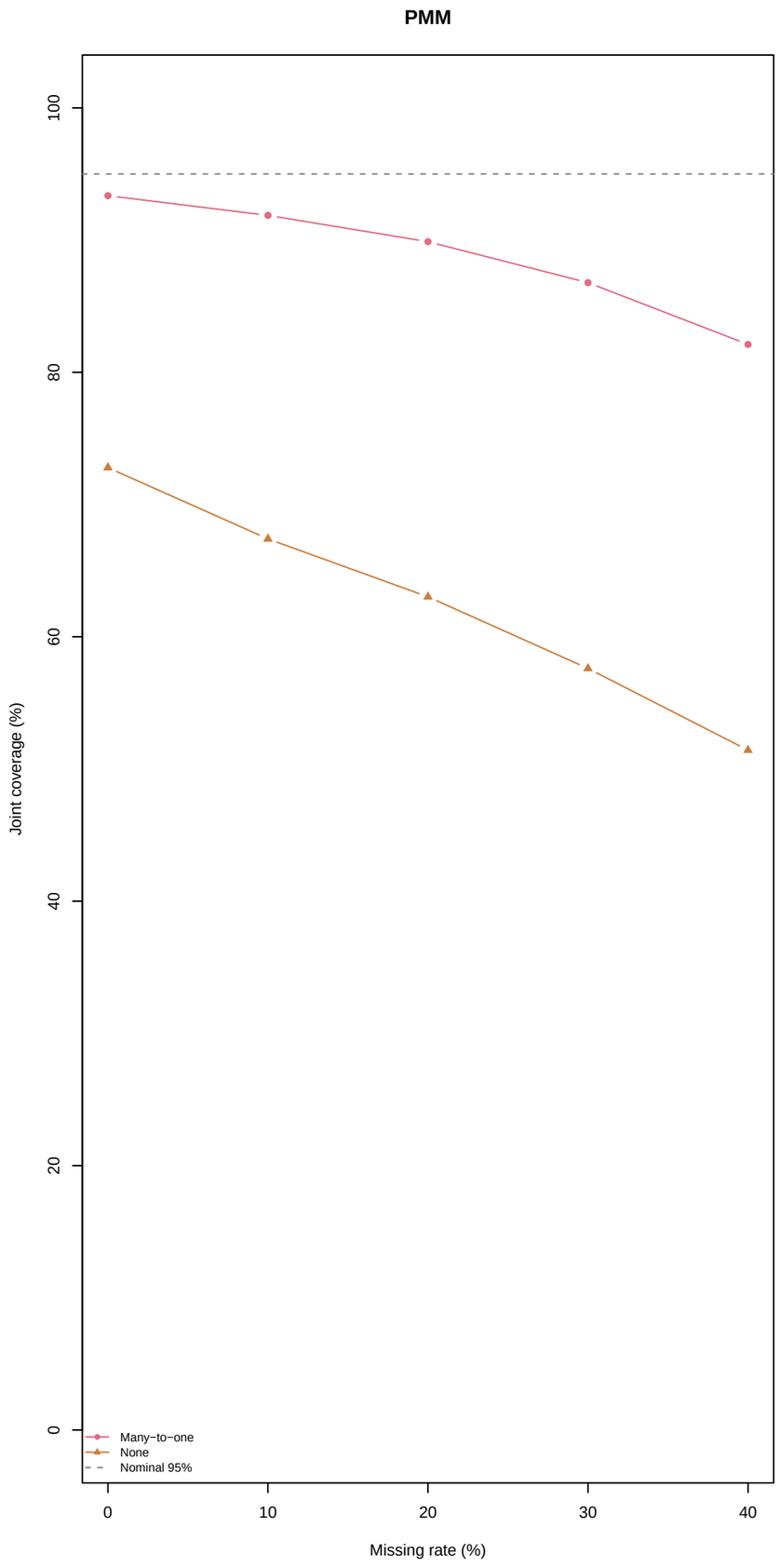



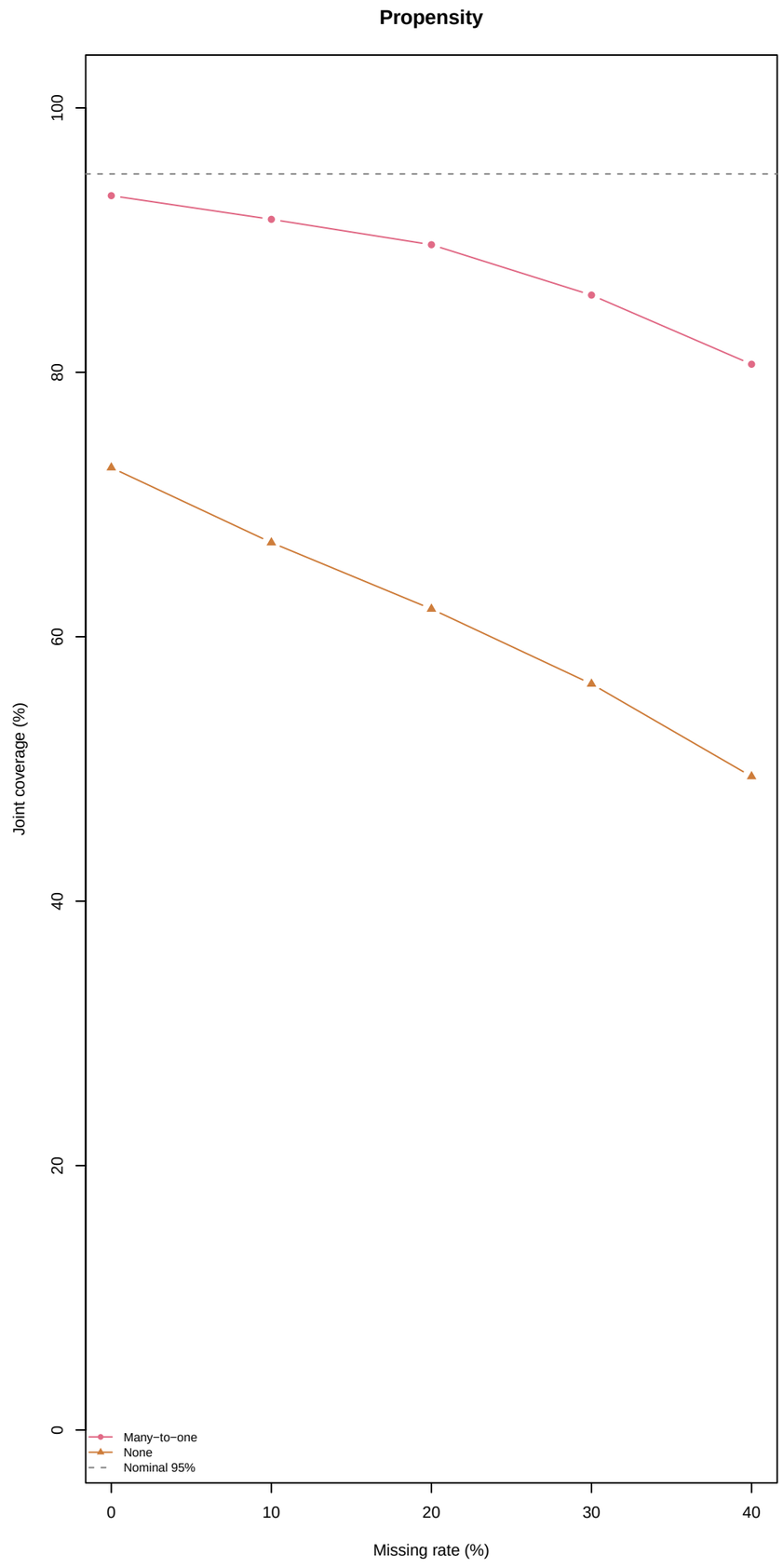



# Joint coverage: Non−shrinkage

Scenario I: all means are 0 for T0,T1,...,T10

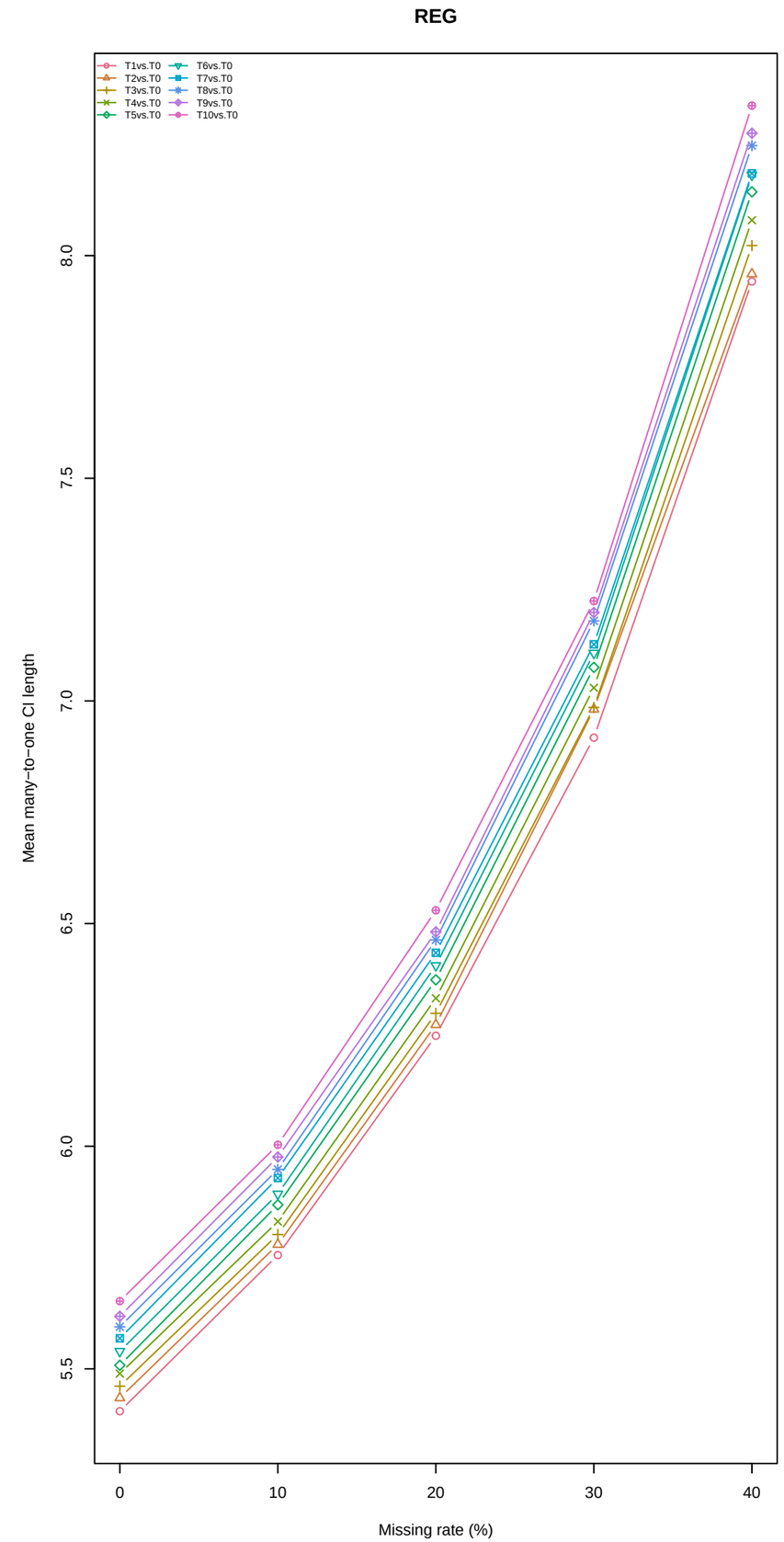


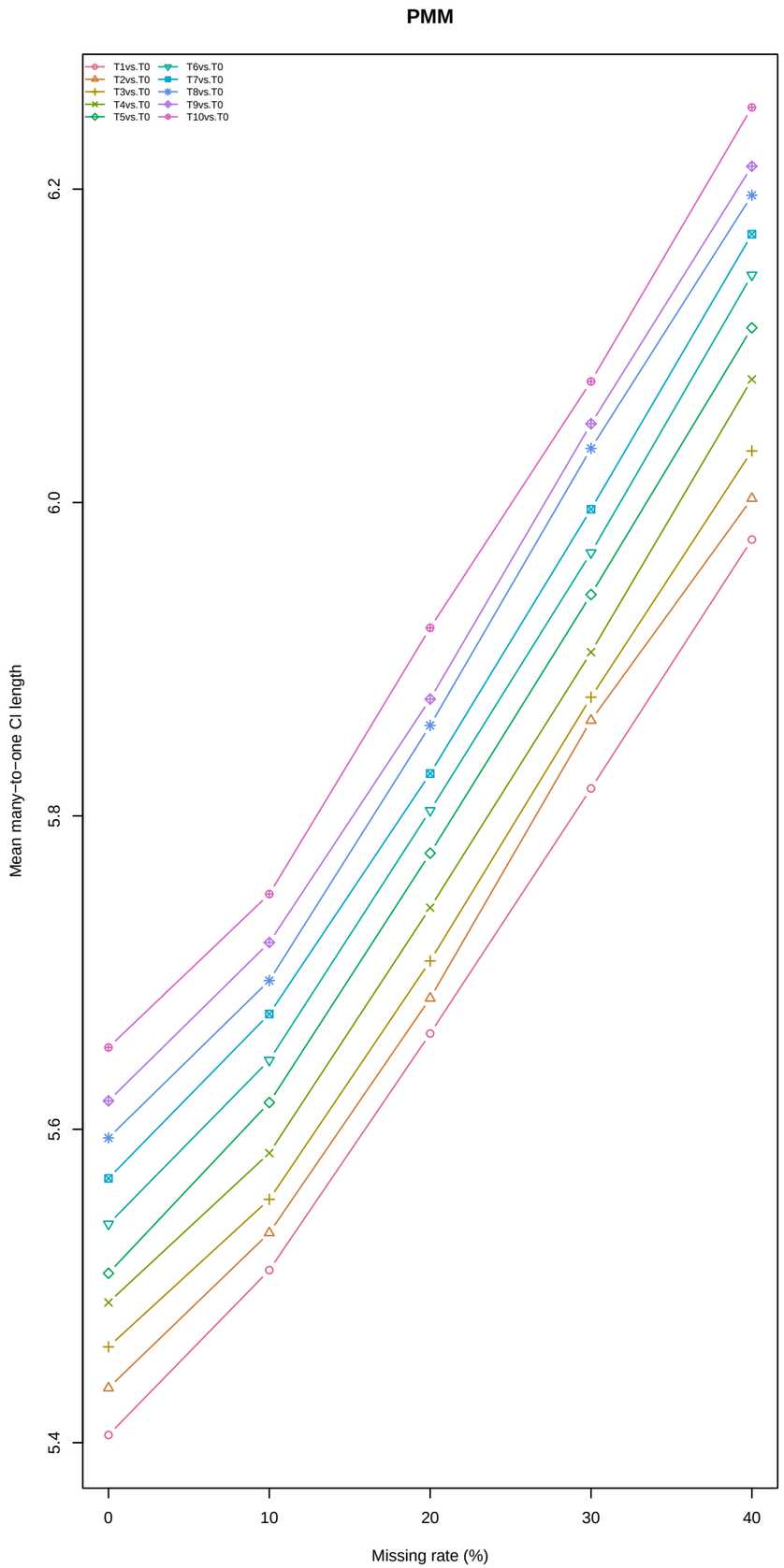



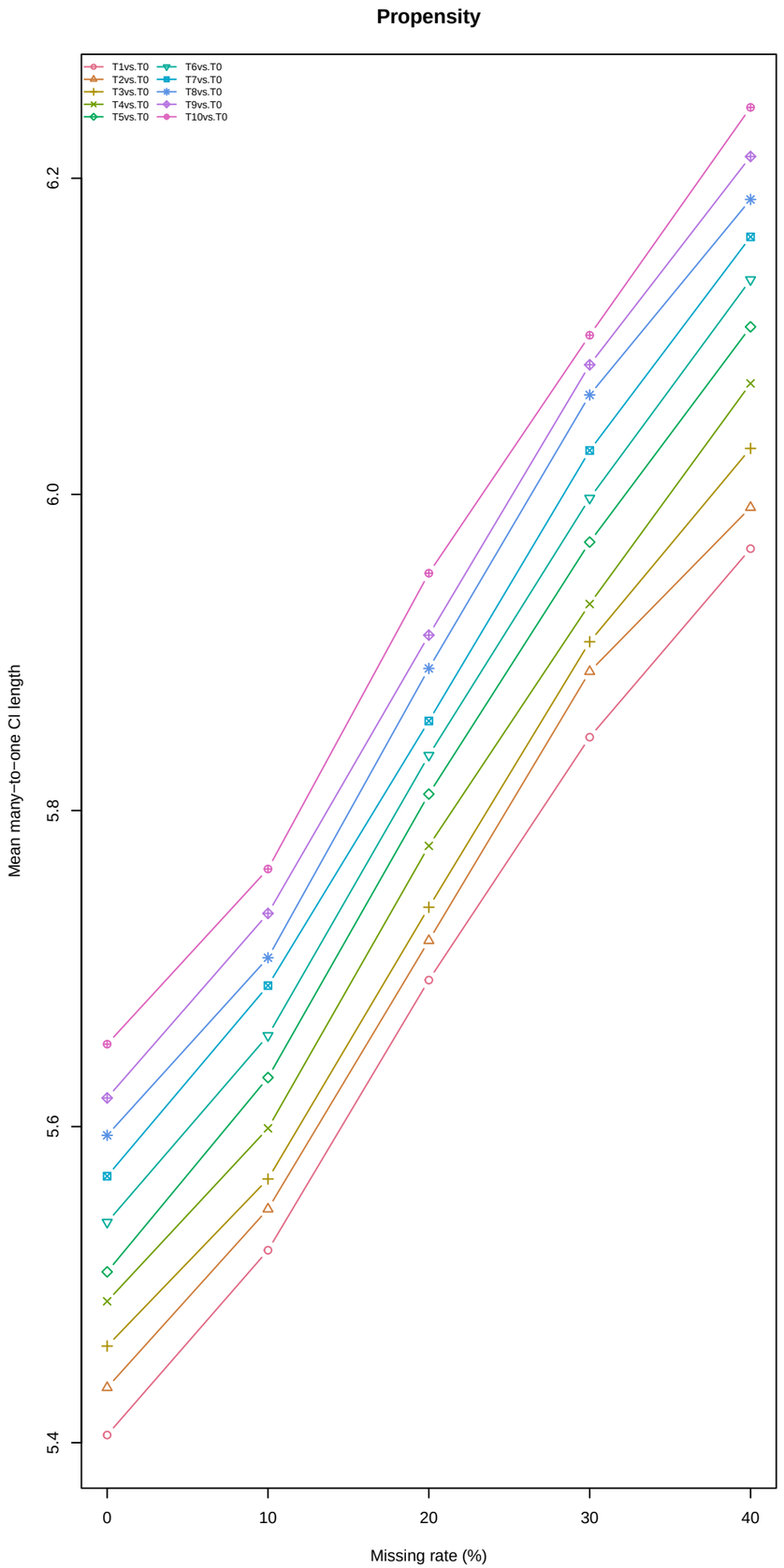



# Interval lengths: Non−shrinkage

Scenario I: all means are 0 for T0,T1,...,T10

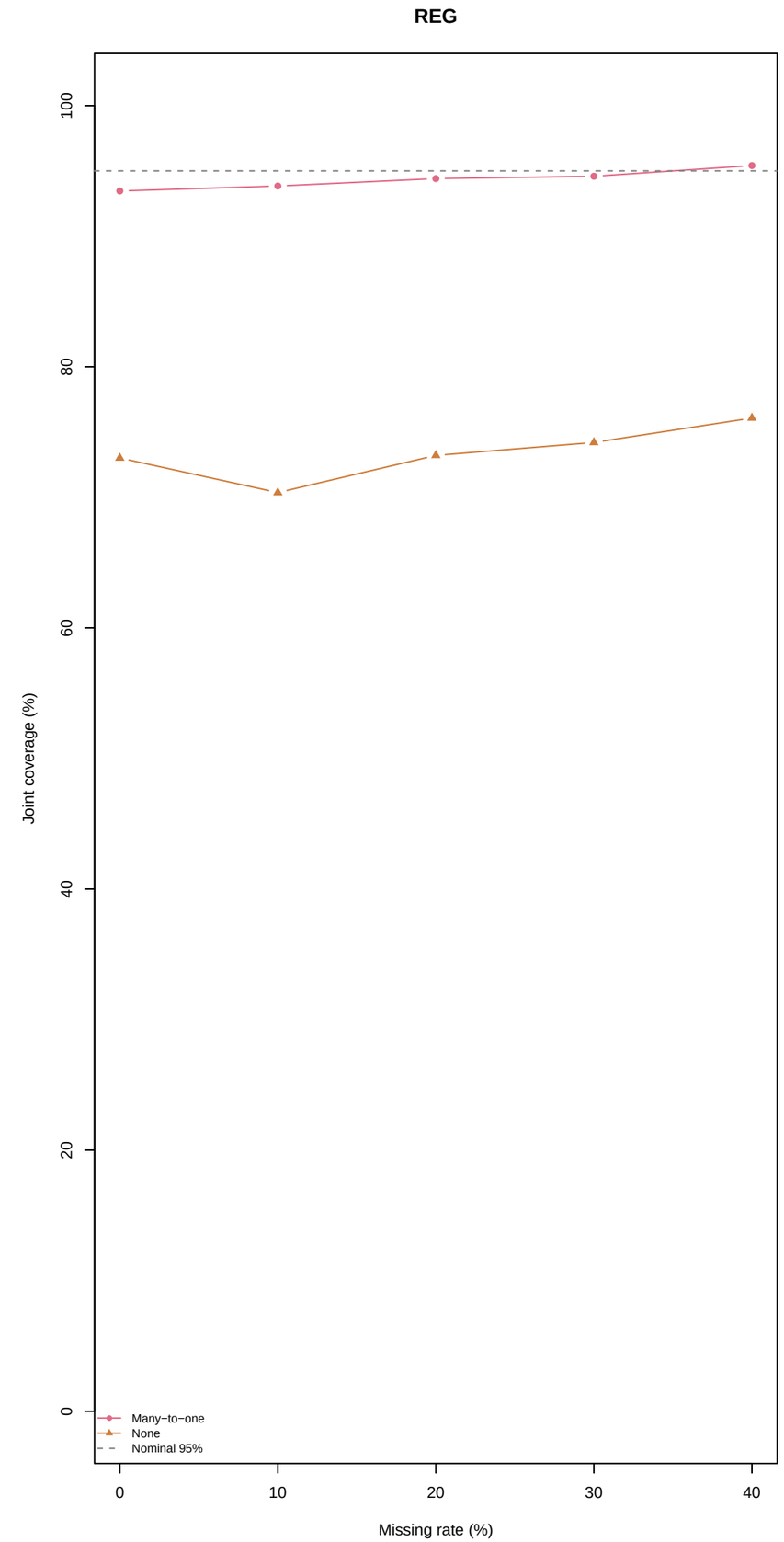


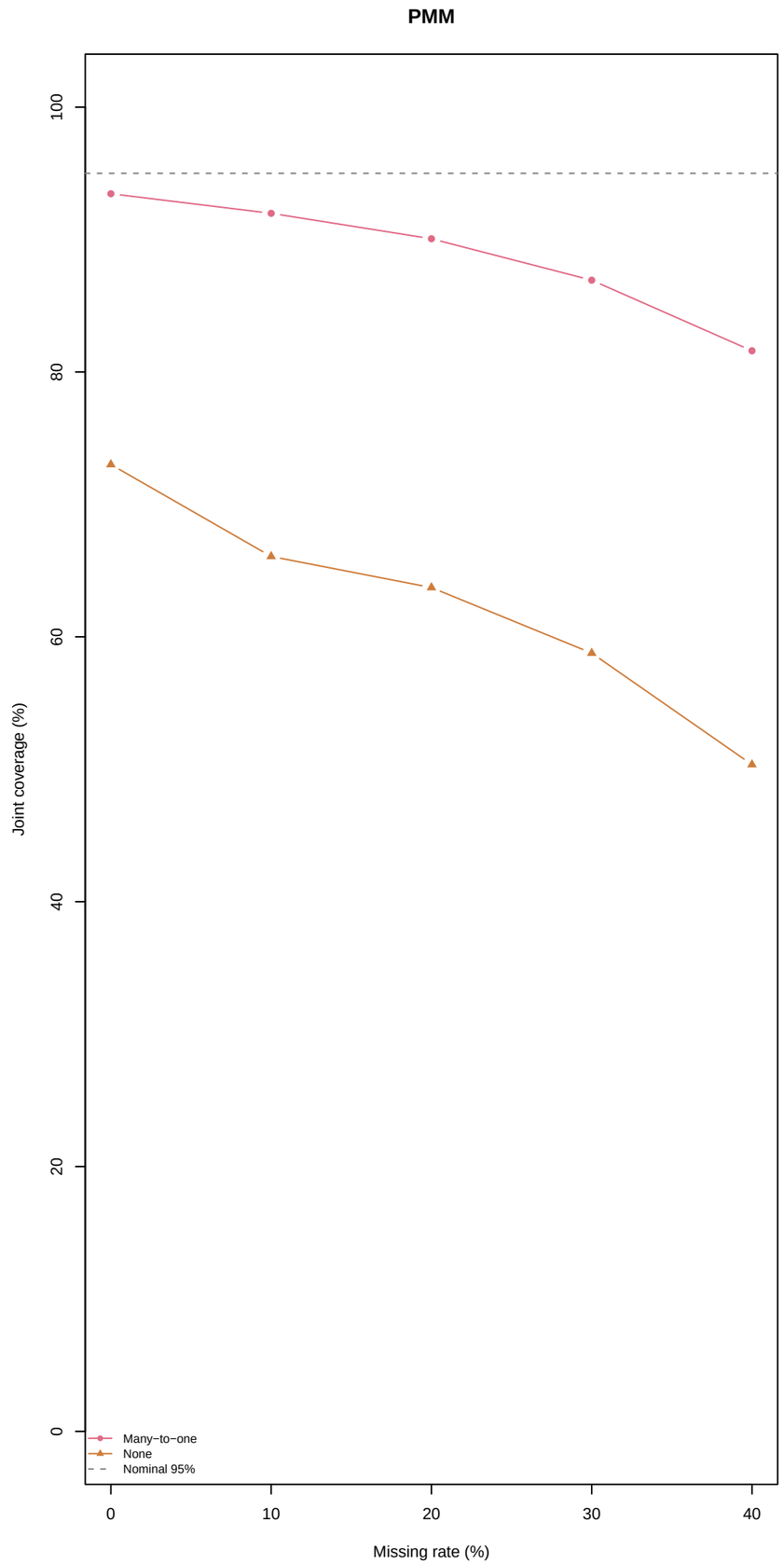



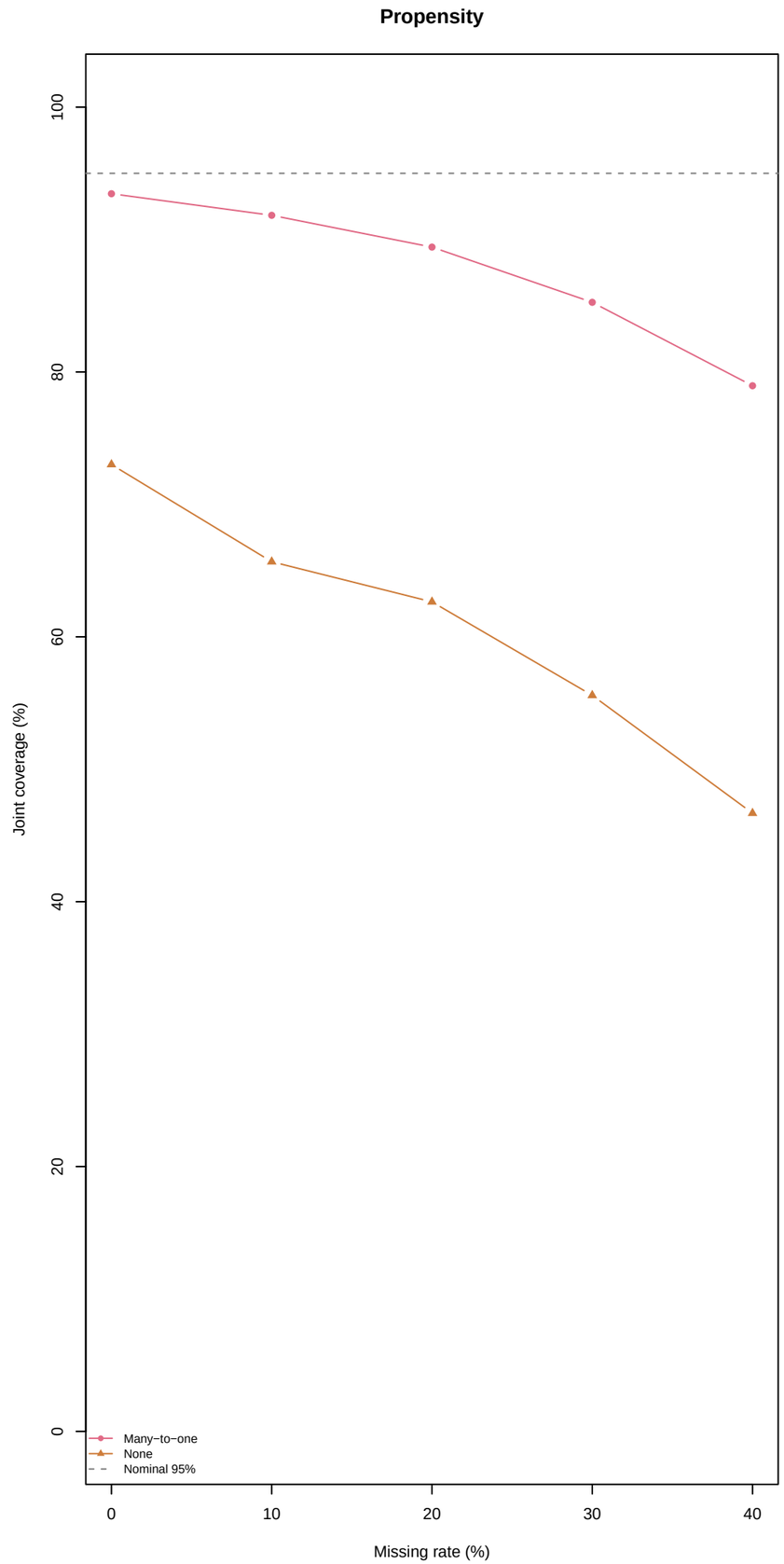



Joint coverage: Non−shrinkage

Scenario II: mu0=0, mu1=1.5, mu2=2.0, ..., mu10=6.0



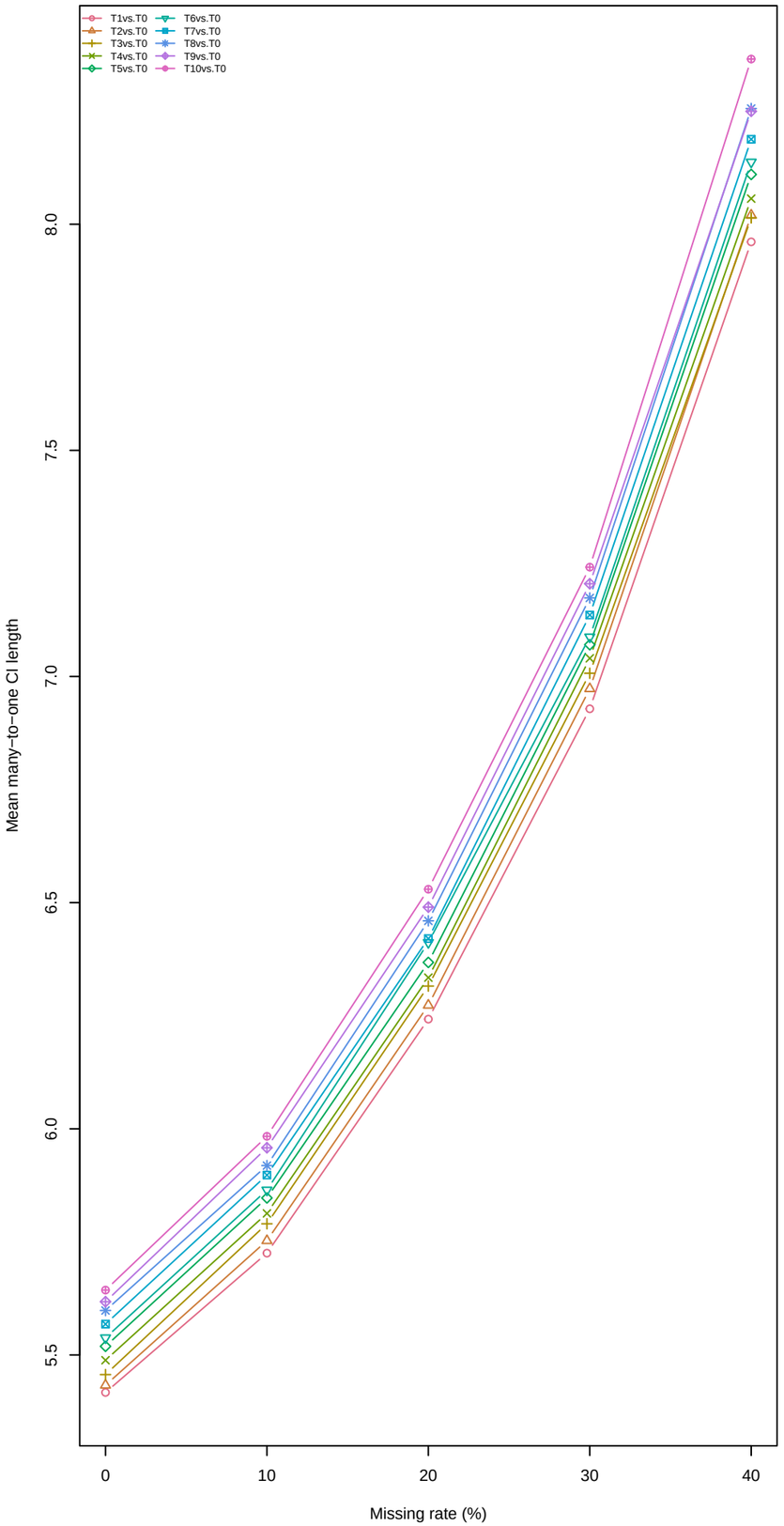



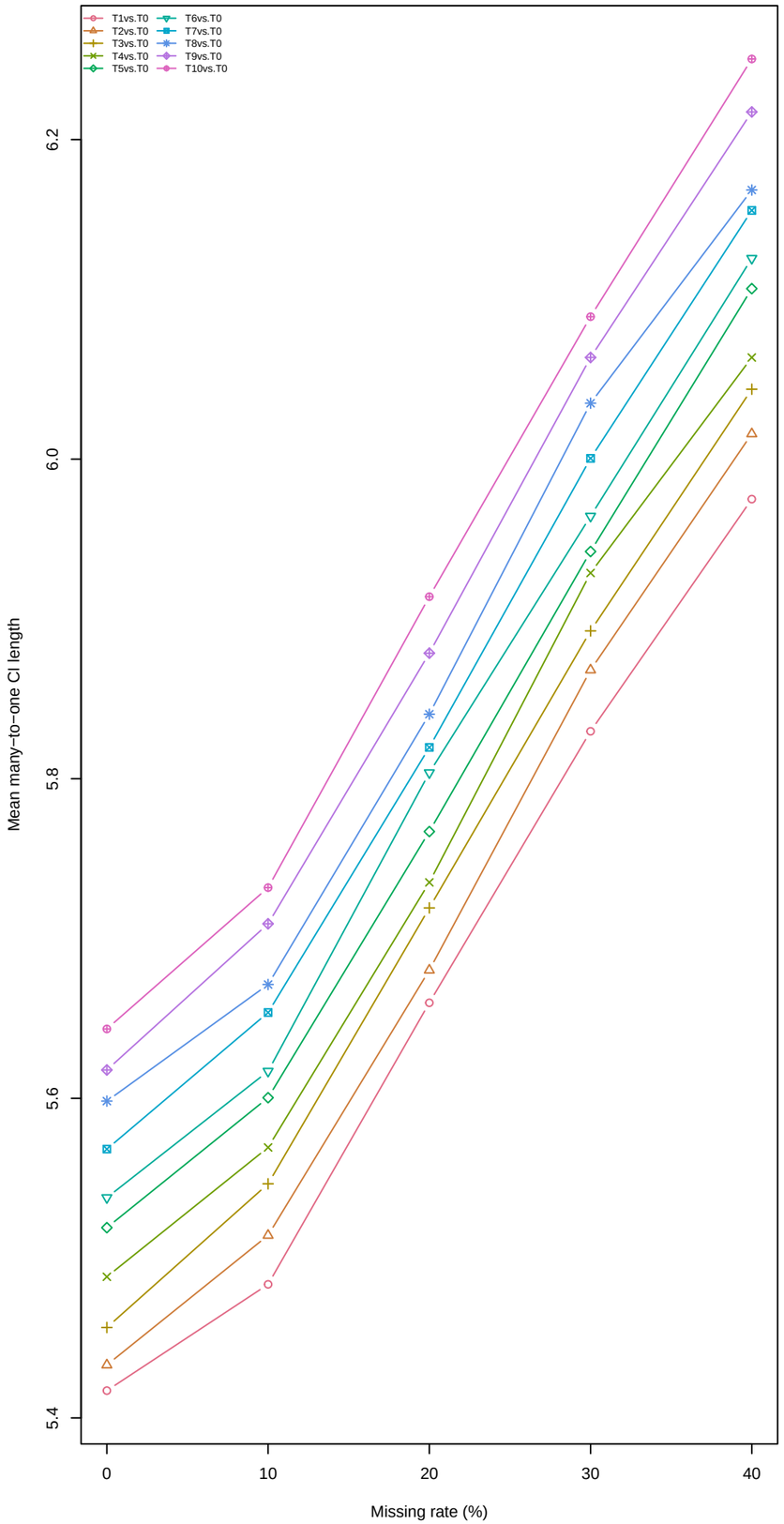
# Interval lengths: Non−shrinkage

Scenario II: mu0=0, mu1=1.5, mu2=2.0, ..., mu10=6.0

## REG




## PMM




## Propensity

